

The Alberta Buck - Notes Flow (v2)

How One BUCK Transfer Becomes N Unlinkable Notes

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One `Notes.mint` transaction takes 1,200 BUCK from Alice and splits it, atomically, into six notes of denominations [1000, 100, 25, 25, 25, 25] – and the public chain learns nothing about who received what. This document walks that example end to end, executably: the SNARK-verified batch mint that folds the note commitments into the on-chain Merkle tree, the off-chain delivery and offline verification of each recipient’s opening, and the three *Identity-M* redemption flows – `spendCoupledA2` (addressed, private issuer), `spendCoupledA1` (addressed, public issuer), and `spendCoupledB1` (bearer, public issuer) – by which a note is deposited by proving, in zero knowledge, both *which Identity may spend it* and *that the Identity is a registered member* of the on-chain accumulator.

The walkthrough demonstrates five concrete guarantees – atomic value conservation, recipient-blinded commitments, per-note disclosure to recipients only, offline receipt reconstruction from the transaction ID alone, and soundness against an arbitrarily hostile issuer-side EVM and prover – and audits the privacy surface step by step: each party can *name* its counterparty under compulsion, while no third party can name either, or link mint to spend. Every cryptographic step is exercised by the shipped `alberta_buck.wallet` code in one seeded, deterministic session, with the output captured in place. (PDF, Text)

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1 Introduction

This document is the *narrative and executable* companion to alberta-buck-notes.org (architectural overview; together they are the two living Notes references) and alberta-buck-ethereum.org (Solidity surface). It walks one concrete example – Alice mints a 1,200-BUCK transfer into six notes of denominations [1000, 100, 25, 25, 25, 25] – end-to-end through the SNARK-verified batch-mint pipeline, and then walks the *inverse* flows: A2 deposit (Bob redeems his identity-targeted note with one folded proof) and B1 deposit (Carol redeems a bearer note while the issuer learns who cashed it). Every actor, every artifact, and every check is named.

The mint pipeline uses SNARK-verified batch minting: the circuit takes `oldRoot` as a public input, folds N commitments into the rolling tree using witness sibling paths, and emits a SNARK-attested `newRoot` for the contract to record. On-chain mint cost is essentially constant in N – $\sim 380K$ plus ~ 500 gas per leaf of calldata.

The spend’s authority axis is the KYC-bound Identity, not an account. The mailbox is a separate receiving key $pk_{\text{recv}} = kG$, derived from the wallet seed and not from the identity scalar, because that scalar is handed to every counterparty and so cannot also open the recipient’s notes. Addressed spends (A1, A2) are one Groth16 proof over one witness (`deposit_fold_a1.circom`, `deposit_fold_a2.circom`): k opens the note, the deposit account’s credential opens to the named Identity, and a registered leaf $\text{Poseidon}(m, k, \text{salt})$ binds the two, on a path to the posted root. A2 adds the membership of the decrypted issuer Identity under a second salt, the one the issuer ships. B1 is not that circuit. Its two claims rest on one secret, so it keeps the depositor-binding sigma and pairs it with `identity_membership_b1.circom`, whose blind is a multiple of H_{PEDERSEN} , hashed to the curve so the discrete log is unknown. `Notes.spendCoupled{A1,A2}` verify that one proof and nothing else: the slot holding the gate cannot be cleared, and an addressed spend with it unset reverts before the spend SNARK is even checked. There is no weaker path to fall back to, which is the property the whole section rests on. The dev setups are not ceremonies.

The cryptographic walkthroughs in this document are *executable*: the Python blocks run against the shipped `alberta_buck.wallet` modules in one seeded, deterministic session, and their printed output is captured below each block.

The aim is to show, by walking the path, that the system delivers five concrete guarantees:

1. **Atomic conservation** – if the transaction succeeds, the BUCK balance leaves Alice exactly once and matches the public total of the new notes. If the proof fails, no BUCK moves and no commitments are appended.
2. **Issuer-bound, recipient-blinded artifacts** – the public commitments name the issuer (`msg.sender`) but reveal nothing about per-note values or recipients.
3. **Per-note disclosure to recipients only** – each recipient receives an *opening* off-chain that lets them reconstruct exactly their own note’s value and the issuer’s verifiable identity, plus the fact that N siblings exist whose hidden values sum to `totalFace` – and nothing more.
4. **Offline receipt reconstruction** – given only the mint transaction ID, anyone holding an opening can independently verify the note against the chain, with no help from Alice or any indexer.
5. **Adversarial-prover soundness** – even if Alice runs a maliciously patched EVM and an arbitrarily hostile witness generator, the honest validators executing the deployed `Notes.sol` and `MintGroth16Verifier.sol` reject every batch that does not satisfy the mint relation. The on-chain truth is what governs.

The one structural caveat (the pinned-N dispatch set under `mint_batch.circom`) is addressed at the end and cross-referenced into the proofs and implementation docs.

2 Cast of Characters

The example uses concrete names so the message flow is easy to track:

- **Alice** – the issuer. She holds 1,200 BUCK and wants to break it into six notes. She is registered with the IdentityRegistry so that her `msg.sender` address resolves to a cryptographically attested identity.
- **Bob, Carol, Dave, Eve, Frank, Grace** – six recipients. Bob will get the 1,000-BUCK note; Carol the 100-BUCK note; Dave/Eve/Frank/Grace one 25-BUCK note each. Recipients do *not* need to be on-chain known to Alice – the recipient identity binding is a field inside the commitment, not an on-chain registration step at mint.
- **The honest EVM** – every Ethereum validator running stock client software. This is the only EVM whose verdict matters. It executes the deployed `Notes.sol`, `MintGroth16Verifier.sol`, `MintVerifierAdapter.sol`, and `Buck.sol` bytecode.
- **Alice’s hostile EVM (Eve-EVM)** – a fictional entity Alice can run locally. She may patch any contract logic she wants, run any witness generator, and produce any proof bytes. The point of the design is that nothing she runs locally affects what the honest EVM accepts.
- **A passive observer (Mallory-the-watcher)** – someone scanning the public chain after the fact, trying to learn *which* note went to *whom* for *how much*. Mallory has full chain history and unlimited compute.

3 The Promise

By the end of the mint transaction Alice will have:

- A single Ethereum transaction `T_mint` on-chain whose effects are:
 - 1,200 BUCK debited from Alice and credited to the `Notes` pool address.
 - A batch of `N*` opaque `uint256` commitments (here `N* = 16`: six live plus ten dummies) folded into the note Merkle tree, moving the root from `oldRoot` to the SNARK-attested `newRoot`, and the tree size from `nextLeafIndex` to `nextLeafIndex + N*`.
 - One `Minted(issuer=Alice, totalFace=1200e18, startIndex=K, count=N*, newRoot)` event; no per-leaf `Appended` events – the commitments themselves live in the transaction calldata.
 - One `Transfer(Alice, Notes, 1200)` ERC-20 event.
 - One Groth16 proof (256 bytes, shape-independent of `N`) in calldata, accepted by the on-chain per-N verifier.
- Six off-chain artifacts, one per *live* recipient, each containing the *opening* of exactly one commitment plus the mint-transaction reference. The ten dummy openings are discarded or retained privately by Alice as anonymity- set filler.

The key invariant: looking at the chain alone, Mallory sees that Alice escrowed 1,200 BUCK and folded N^* hashes into the tree. She cannot tell which hash hides which value, who any recipient is, or even whether two of the recipients are the same person – or which entries are live notes vs. zero-value padding. Looking at his off-chain artifact, Bob sees his 1,000 BUCK is real, was minted by Alice in T_{mint} , and that the *live* siblings among the other fifteen have hidden values summing to exactly 200 BUCK (the rest are $v = 0$ padding he cannot distinguish from live notes by chain inspection alone) – but he does not learn what those values are or who holds them.

4 What’s Inside a Note (Conceptually)

A "note" is just a five-tuple known by Alice and the recipient:

```
opening_i = ( flavor_i, v_i, rho_i, idHash_i, predicate_i )
cm_i       = Poseidon5( flavor_i, v_i, rho_i, idHash_i, predicate_i )
```

- **flavor_i** – which note variant (A1 identified cheque, A2 private cheque, B1 bearer cheque); see Notes for the menu.
- **v_i** – the face value in BUCK base units (18-decimal scaled integer).
- **rho_i** – a fresh, uniformly random nonce. This is the source of unlinkability: swapping **rho_i** changes **cm_i** to a uniformly fresh hash even if every other field is identical.
- **idHash_i** – a Poseidon-hashed handle on the recipient’s identity (e.g. a Poseidon-hash of the recipient’s registered ElGamal credential, or zero for a pure bearer note).
- **predicate_i** – a domain-separating tag for any optional spending condition (escrow, time-lock, etc.). Zero in the simplest case.

The Poseidon-5 hash gives the commitment two cryptographic properties that make the whole flow work:

- **Hiding.** Without **rho_i**, the commitment leaks no information about the other fields. Mallory sees **cm_i** and gains nothing about **v_i**, **idHash_i**, or **flavor_i**.
- **Binding.** No prover, hostile or honest, can find two distinct openings hashing to the same **cm_i** (collision-resistance of Poseidon on BN254, modelled as a random oracle in alberta-buck-proofs.org Part IV).

The on-chain contract stores only **cm_i**. Openings live exclusively off-chain.

5 The Mint, Step by Step

5.1 Step 0: Pre-flight (Off-Chain, In Alice’s Wallet)

Alice’s wallet has done all the slow work before any transaction is broadcast:

- She owns 1,200 BUCK on-chain.

- She has decided on the denomination split: [1000, 100, 25, 25, 25, 25]. $N=6$. The deployed mint pipeline pins a small set of batch sizes – recommended {16, 128, 1024} – one circom circuit and one verifier per pin. Alice’s wallet rounds her requested split up to the smallest pin $N^* \geq 6$, here $N^* = 16$, and pads with $16 - 6 = 10$ "blinded zero" openings whose $v_i = 0$. The padding leaves cost the same SNARK constraints as live notes but contribute zero to `totalFace` and are spendable for zero BUCK; in effect they are anonymity-set filler whose openings Alice may discard. The narrative below tracks the six live notes and ignores the ten dummies (they show up in the on-chain commitments[] vector alongside the live ones, indistinguishable to Mallory).
- She has chosen an `idHash_i` for each recipient (typically a Poseidon hash of the recipient’s IdentityRegistry-bound ElGamal credential, derived once at out-of-band introduction time).
- Her CSPRNG produces a fresh `rho_i` nonce per leaf – live or padded.

She now has 16 opening tuples in memory: 6 live, 10 dummies.

5.2 Step 1: Build Commitments

For $i = 0..15$ (six live, ten dummies) her wallet computes

```
cm_i = Poseidon5( flavor_i, v_i, rho_i, idHash_i, predicate_i )
```

This is pure off-chain Poseidon arithmetic in `alberta_buck.wallet.notes`. Sixteen `uint256` field elements drop out. Their order in the eventual on-chain array is fixed by Alice’s local list – the contract preserves whatever order she submits, and the SNARK assumes leaves land in the tree at indices `[nextLeafIndex, nextLeafIndex+1, ..., nextLeafIndex+N-1]` in submission order.

Alice’s wallet must also keep a *local mirror of the rolling tree state*. The Tornado-style Merkle tree the contract maintains is parameterised by the standard `filledSubtrees[d]` for $d = 0..TREE_DEPTH-1$ – the rightmost partially-built node at each level. The wallet snapshots this state from the chain (or replays it from the mint event log) immediately before assembling the proof, then derives, for each of the 16 leaves *in submission order*, the depth-20 sibling sequence the in-circuit insertion will consume. That derivation is the one delicate piece of wallet-side mint code; everything else is straightforward Poseidon arithmetic.

5.3 Step 2: Generate the Batch-Mint Groth16 Proof

Alice’s wallet runs `scripts/snark/prove_mint_batch.js`. The `mint_batch.circom` circuit at her chosen pin $N^* = 16$ takes a witness that covers not just the openings but the tree update itself, because the Merkle insertion lives *inside* the SNARK.

Public signals (circom emits the circuit OUTPUTS first, then the public inputs; arity $2N+4$):

```
issuerMode[i]  # i in 0..N-1: per-leaf PUBLIC=1 / PRIVATE=2 (a circuit OUTPUT,
                # the deterministic flavor projection -- Notes.mint gates on it)
oldRoot        # the live note root she read from chain
newRoot        # the root after folding her N leaves
nextLeafIndex  # the live tree size she read from chain
totalFace      # 1200e18 -- range-bounded to 128 bits
cm[i]          # i in 0..N-1: the commitments, exposed DIRECTLY (not a batch hash)
```

The commitments are exposed as public inputs directly (no batch digest): each input costs $\sim 6K$ gas in the EVM verifier, cheaper than recomputing a keccak over calldata on-chain, and far cheaper

than an in-circuit Poseidon over N elements. (A2 / private-issuer batches use a *separate* circuit, `mint_batch_a2`, that exposes each leaf’s `eIss` instead of `issuerMode`; see *The Redemption* and the Decryptability companion.)

Private witness (per-leaf and per-level):

```
For i in 0..N-1:
  flavor[i], v[i], rho[i], idHash[i], predicate[i]    # the opening
  pathSiblings[i][d] for d in 0..TREE_DEPTH-1        # 20 siblings per leaf
```

The `pathSiblings` for leaf i are derived by the wallet from its local `filledSubtrees` mirror plus the previously-folded leaves of *this same batch* (leaves $0..i-1$ sit at known positions and contribute their Poseidon hashes to i ’s path on the right). The wallet does this work; the chain consumes only `oldRoot`, `newRoot`, and the calldata `cms[]`.

In-circuit constraints:

1. **Per-leaf opening.** For each i , `cm[i] == Poseidon5(flavor[i], v[i], rho[i], idHash[i], predicate[i])`. N Poseidon-5 evaluations, N equality constraints.
2. **Range bounds.** Each `v[i]` and `totalFace` passes through `Num2Bits(128)`.
3. **Sum conservation.** `sum(v[0..N-1]) == totalFace`.
4. **Rolling Merkle insertion.** The circuit threads `oldRoot` and `nextLeafIndex` through N successive single-leaf insertions – the standard Tornado-style incremental-tree fold, performed in-circuit: at each step it folds `cm[i]` up the tree using `pathSiblings[i][*]` and `Switcher(...)` per level, advances the rolling state, and finally asserts the resulting root equals `newRoot`. $N \times \text{TREE_DEPTH}$ Poseidon-2 hashes ($\sim 4.3\text{K}$ constraints per leaf).
5. **Flavor range + issuer mode.** Each `flavor[i] \in \{A1=1, A2=2, B1=3\}`, and `issuerMode[i]` is the deterministic projection ($A2 \rightarrow \text{PRIVATE}$, else `PUBLIC`), exposed as a public output. This is what makes a bearer leaf provably public-issuer (and a private-issuer bearer note unmintable – see the Decryptability companion, *Why B2 is impossible*).

Constraint counts work out to roughly **21,800 R1CS per leaf** (opening + range-bound + dual Merkle walk: one path consumed to attest `oldRoot`, one to advance `newRoot`); the shipped pinned- N set measures 21,914 R1CS at $N=1$, 348,705 at $N=16$, and 697,281 at $N=32$. The doubling per pin holds linearly through the projected sizes ($N=128 \sim 2.8\text{M}$ R1CS, $N=1024 \sim 22\text{M}$). At $N=16$ the 2^{19} PTAU and ~ 6 s prover wall on a dev laptop are comfortable; at $N=128$ it is rapid-snark-and-minutes; at $N=1024$ it is hours plus a multi-GB zkey. The `scripts/snark/setup.sh` ceremony emits one verifier per pin.

The witness is processed by snarkjs’s Groth16 prover. Tens of seconds to minutes later the wallet has a fixed-size proof object (`pi_A: G1, pi_B: G2, pi_C: G1`), encoded as ABI-static (`uint256[2], uint256[2][2], uint256[2]`) – exactly 256 bytes regardless of N .

Observation. The Groth16 proof reveals *nothing* about the witness. Soundness guarantees that no `proofBytes` can satisfy the verification equation unless a witness exists; zero-knowledge guarantees that the verification equation reveals nothing about which witness was used. Mallory, even with the proof in hand, cannot back out `v[i]`, `rho[i]`, `idHash[i]`, `flavor[i]`, `predicate[i]`, or the per-leaf Merkle paths – only `oldRoot`, `newRoot`, `nextLeafIndex`, `totalFace`, the `cm[i]`, and the per-leaf `issuerMode[i]` (flavor-class: public vs private, no identity) leak.

5.4 Step 3: Submit Notes.mint On Chain

Alice now broadcasts a single transaction `T_mint` calling

```
// Minting is gated-only: a nameable issuer Identity is bound to every batch.
// Alice (a public Corporate Identity here) uses the PUBLIC overload:
notes.mint(
    proofBytes,
    oldRoot, newRoot, nextLeafIndex,
    1200e18, // totalFace
    [cm_0, cm_1, ..., cm_15], // cms.length == N* (here 16)
    [PUBLIC, ..., PUBLIC], // issuerMode (a SNARK public input of mint_batch)
    issuerSig // Schnorr by Alice's registered key over keccak(cms)
);
// A private (A2) issuer uses the other overload instead:
// notes.mint(..., cms, issuerMode, a2Bindings) // routes the proof to mint_batch_a2
// // and ties each leaf's eIss
// There is no unbound mint path -- an unnameable note cannot be created.
```

The honest EVM walks `Notes.mint` (the `PUBLIC` overload):

```
// Stale-state guards + field bounds (_verifyMintOrRevert).
require(cms.length > 0, "Notes: empty mint");
require(issuerMode.length == cms.length, "Notes: issuerMode/cms length");
require(oldRoot == roots[currentRootIndex], "Notes: stale oldRoot");
require(nextLeafIndex == self.nextLeafIndex, "Notes: stale nextLeafIndex");
require(newRoot != 0 && newRoot < FIELD_R, "Notes: bad newRoot");

// SNARK verify: the proof binds issuerMode + cm[] + the roots. The
// MintVerifierAdapter dispatches by cms.length to the per-N Groth16 verifier;
// public-signal vector is [issuerMode[0..N], oldRoot, newRoot, nextLeafIndex,
// totalFace, cm[0..N]] (outputs lead), arity 2N+4.
require(mintVerifier.verifyMint(
    proofBytes, issuerMode, oldRoot, newRoot, nextLeafIndex, totalFace, cms
), "Notes: bad mint proof");

// Public-issuer binding (gated-only): every leaf must be PUBLIC-mode,
// msg.sender must be a registered PUBLIC Identity, and issuerSig must be a valid
// Schnorr over keccak256(cms). (The PRIVATE-mode overload instead routes the
// proof to mint_batch_a2 and verifies one A2 re-encryption binding per leaf,
// tying each committed leaf's eIss.)
require(identityRegistry.isPublicIdentity(msg.sender),
    "Notes: public-mode leaf needs public issuer");
require(identityRegistry.verifyIssuerSchnorr(
    msg.sender, keccak256(abi.encodePacked(cms)), issuerSig),
    "Notes: bad issuer binding");

// Pull BUCK only after the proof is accepted, then advance (_advanceAndPull).
require(buck.transferFrom(msg.sender, address(this), totalFace),
    "Notes: transfer failed");
currentRootIndex = (currentRootIndex + 1) % ROOT_HISTORY_SIZE;
roots[currentRootIndex] = newRoot;
self.nextLeafIndex += cms.length;
noteFaceSum += totalFace;

// Single batch event; cms[] live in calldata.
emit Minted(msg.sender, totalFace, /*startIndex=*/ nextLeafIndex,
    /*count=*/ cms.length, newRoot);
```

Five things happen in strict order, and **all of them are atomic** (one transaction, one EVM state transition):

a. **Stale-state guards.** The contract asserts that `oldRoot` equals the current ring-buffer head and `nextLeafIndex` equals the live tree size. Either fails if another mint landed since Alice's wallet snapshotted state (more on this under **Prover Contention and the Rollup Race**

below). b. **Calldata binding.** The `cms[]` are exposed **directly** as Groth16 public inputs (no batch digest), alongside the per-leaf `issuerMode[]` the circuit emits, so any tampering with `cms[]` or `issuerMode[]` en route from the prover to the chain mismatches the SNARK and the verify call rejects. c. **SNARK verify.** The `MintVerifierAdapter` dispatches by `cms.length` to the per-N verifier (one circuit per pin, one verifier contract per pin), unpacks the 256-byte `proofBytes` into Groth16 form, builds the $2N+4$ -entry public-signal vector `[issuerMode[0..N], oldRoot, newRoot, nextLeafIndex, totalFace, cm[0..N]]` (circuit outputs lead), and invokes the appropriate `MintBatchN $\{N\}$ Groth16Verifier`. If the verifier returns false, the entire transaction reverts – nothing is written, no BUCK moves. d. **BUCK escrow.** `buck.transferFrom` debits `totalFace` from Alice and credits the Notes pool. Allowance-driven pull, gated by Alice’s prior `approve(notes, 1200e18, E_notes, pi_CP)`. This CP-bound `approve` is mandatory: Alice is a private EOA, and mutual decryptability requires a per-pair CP fragment for every private party (the `identityHash` fallback is only valid for Public-Identity contracts). The pool was bound under a Public Identity (`bindContract(notes, pk, E, true)`), so the pool side falls back to `_identityHash(notes)` without requiring a fragment from the pool. The Notes operator decrypts Alice’s CP ciphertext from the `IdentityExchange` event under subpoena. e. **Root advancement.** The contract writes `newRoot` into the next ring-buffer slot, advances `currentRootIndex` and `nextLeafIndex`, accumulates `noteFaceSum`, and emits a single `Minted(issuer, totalFace, startIndex, count, newRoot)` event. The new root immediately becomes the live root that the next minter must read.

If any of (a)-(e) fails – stale root, stale leafIndex, bad proof, insufficient BUCK, missing approval – the entire transaction reverts and the chain state is indistinguishable from no submission at all. **Alice gets one shot per transaction;** she cannot retry a partial state.

Note on duplicate commitments. The contract carries no `commitmentExists[cm]` duplicate check – not because duplicates are harmless, but because they are *economically self-punishing*: if Alice submits the same opening twice (whether across two batches or twice in one), she pays `totalFace` on each occurrence, the leaves land at distinct tree indices, but the spend nullifier `Poseidon3(rho, idHash, 4242)` is *deterministic in the opening*. After the first spend nullifies `cm`, the second copy is unspendable forever. Net result: Alice burns BUCK she could have kept. The chain’s job is value conservation, not saving Alice from a buggy `rho` RNG.

5.5 Step 4: Off-Chain Note Delivery

After `T_mint` confirms, Alice transmits each live opening to its intended recipient by any out-of-band channel: an encrypted message, a printed QR code, a USB stick. The payload to recipient `i` is:

```
note_artifact_i = {
  txHash:    T_mint,                # Ethereum transaction ID
  leafIndex: startIndex + i,        # absolute position in the commitment vector
  cm:        cm_i,                  # for cross-checking against calldata
  opening:   ( flavor_i, v_i, rho_i, idHash_i, predicate_i ),
  meta:      { issuer: Alice's IdentityRegistry handle, batchSize: N*, ... },
}
```

The chain itself never sees `opening`. `cm_i` is already folded into `newRoot`; the artifact carries it explicitly only as a convenience so the recipient does not have to recompute it before locating it in the batch’s calldata `cms[]`.

The ten dummy openings Alice generated to pad to the pinned $N=16$ circuit receive no artifact; they are discarded (or, at Alice’s option, retained privately as unspendable anonymity-set filler).

From the recipient’s perspective the presence of dummies is invisible: Bob’s artifact only names his own leaf, and the batch total `totalFace = 1,200 BUCK` is exactly the sum of the live `v_i`.

5.6 Step 5: Recipient Verification (Offline, Independent)

Bob receives his artifact for the 1,000-BUCK note. Without trusting Alice and without any help from an indexer, he runs:

1. **Recompute and check.** Bob computes `cm_check = Poseidon5(flavor, 1000e18, rho, idHash, predicate)` and asserts `cm_check == cm`. If this fails, Alice lied about the opening; Bob discards the artifact and demands a real one. (Binding of Poseidon makes substitution impossible.)
2. **Look up the mint transaction.** Bob reads `T_mint` on-chain, parses the call data into `(proofBytes, oldRoot, newRoot, nextLeafIndex, totalFace, cms[])` and reads `msg.sender` from the transaction envelope.
3. **Confirm `msg.sender` is Alice.** Cross-check `msg.sender` against the IdentityRegistry: it must resolve to Alice’s PS-credentialed identity. If Bob wanted an A1 identified cheque, the issuer field of his opening must match.
4. **Confirm calldata membership.** Assert `cms[leafIndex - nextLeafIndex] == cm` (where the artifact’s `leafIndex` is the absolute position). The chain’s acceptance of `T_mint` confirms that every entry in `cms[]` was folded into `newRoot` and that `newRoot` was added to the root ring buffer. Bob therefore has chain-anchored evidence that `cm` is officially a commitment in the pool.
5. **Confirm batch totals.** Read `totalFace` from the call data. The chain’s acceptance of `T_mint` is itself a proof that the Groth16 verifier accepted – which means `sum(v[0..N-1]) == totalFace` holds in the witness. Bob does not need to re-verify the proof; the EVM already did. But if he wants belt-and-suspenders, he can re-run `MintGroth16Verifier.verify(proofBytes, [issuerMode[0..N], oldRoot, newRoot, nextLeafIndex, totalFace, cm[0..N]])` locally against the same on-chain bytecode, assembling the public-signal vector from the calldata exactly as the contract does.

After (1)-(5) Bob knows:

- His note encodes exactly 1,000 BUCK. (Step 1.)
- It was minted in `T_mint` by Alice. (Steps 2, 3.)
- It is one of N^* (here 16) commitments folded into the tree by that mint, the others being `cms[] \ cm`. (Step 4.)
- The *live* values among the other siblings sum to exactly $1200 - 1000 = 200$ BUCK (the residual of `totalFace` once Bob subtracts his own `v`); the batch may also contain dummy openings with `v = 0`. (Step 5 + the per-note opening relation: he knows *his* value, the chain attests to the *sum*.)
- He learns nothing else about the other notes – not their individual values, not their recipients, not their flavors, not which are live and which are padding. Each `cm_j` is hiding under a fresh `rho_j` he does not have.

This is the unlinkability promise made concrete: each recipient can pin down their own slice and the batch total, without learning anything about siblings.

6 Why Alice Cannot Cheat (Even With a Hostile EVM and Prover)

Alice runs whatever software she wants on her laptop. She can fork the snarkjs witness generator, patch the circom compiler, run Anvil with a malicious `MintGroth16Verifier` that always accepts. None of it matters, because the truth on the public chain is determined by what *the honest validators* run. Three pillars prop up the soundness story:

6.1 Pillar 1: The On-Chain Verifier Is Fixed Code

`MintGroth16Verifier.sol` (one per pinned N) was emitted from the Groth16 verification key produced by the trusted setup (production ceremony pending; dev-only entropy in `scripts/snark/setup.sh` today). Its bytecode is deployed once and immutable. Validators run whatever bytecode sits at that address; Alice cannot tamper with it without consensus-breaking the chain.

Knowledge soundness of Groth16 means: for every `proofBytes` the verifier accepts on public inputs `[issuerMode[0..N], oldRoot, newRoot, nextLeafIndex, totalFace, cm[0..N]]`, there exists a witness `(flavor[i], v[i], rho[i], idHash[i], predicate[i], pathSiblings[i][*])_i` such that:

- `cm[i] = Poseidon5(flavor[i], v[i], rho[i], idHash[i], predicate[i])` for all `i`
- each `v[i]` and `totalFace` fit in 128 bits and `sum(v[i]) = totalFace`
- folding `cm[0..N-1]` into the tree rooted at `oldRoot`, starting at position `nextLeafIndex`, using `pathSiblings[i][*]` as the right-of-insert context, yields exactly `newRoot`
- each `issuerMode[i]` is the deterministic flavor projection (`A2 → PRIVATE`, else `PUBLIC`) of the witness `flavor[i]`.

Alice is free to *find* such a witness any way she likes – she’s the prover. But she cannot find a witness for a *false* statement; that’s what soundness prohibits.

6.2 Pillar 2: BUCK Escrow Is Inside the Same Atomic Call

The proof check, the BUCK pull, and the root advancement all live in the same `Notes.mint` function. An EVM transaction is all-or-nothing: the only two outcomes are "the entire call’s state changes commit" or "the entire call’s state changes revert".

This means Alice cannot, for instance:

- Verify a proof, then refuse to pay – the verifier acceptance and the `transferFrom` live in the same call, gated by the same `require`.
- Pay 1,200 BUCK but then get away with advancing only part of the tree – `newRoot` is SNARK-attested as the deterministic fold of `all` of `cms[]` onto `oldRoot` at `nextLeafIndex`, and the contract writes it in one store.
- Replay a prior `proofBytes` – the SNARK is bound to `(oldRoot, newRoot, nextLeafIndex, totalFace)` and the `cm[] / issuerMode[]` vectors, and the contract’s stale-state guards reject any submission whose `oldRoot` no longer matches the live root or whose `nextLeafIndex` no longer matches the live tree size.

- Submit the same opening twice *in one batch* to double-spend at spend time – the nullifier is deterministic in (`rho`, `idHash`), so whichever duplicate lands first at spend will consume both. (See the duplicate-commitment note under Step 3.)

6.3 Pillar 3: Public Inputs Are Sealed Into the Proof

The Groth16 verifier consumes the public-signal vector [`issuerMode[0..N)`, `oldRoot`, `newRoot`, `nextLeafIndex`, `totalFace`, `cm[0..N)`]. The proof was generated against the specific values Alice committed to before submission. If Alice tries to submit the proof with a different `totalFace`, a different `newRoot`, a different `nextLeafIndex`, or a permuted or tampered `cms[]`, the verifier rejects.

The `cm[i]` are public inputs to the SNARK directly: the contract passes the calldata `cms[]` straight into the verifier’s public-signal vector, so any mismatch between the commitments the proof sealed and the commitments the transaction carries reverts at verify time.

The refreshed `test/MintVerifier.t.sol` should cover at least these tamper paths, each expected to revert with `Notes: bad mint proof` or `Notes: stale oldRoot` or `Notes: stale leafIndex` or an ABI decode error:

- `totalFace + 1`
- `newRoot` flipped by one bit
- `oldRoot` spoofed to a stale value
- `nextLeafIndex` spoofed to a fictional size
- `cms[0] ^ 1` (commitment tampered in calldata after proving)
- `cms[]` length differs from the dispatched verifier’s pinned `N`
- malformed 256-byte `proofBytes`

Soundness is not a proof of *absence* of attack, but every documented attack surface is closed by a test that triggers a verifier reject.

6.4 Putting the Pillars Together: The Adversarial Walk

Suppose hostile-Alice tries to mint two **contradictory** batches that both claim to be the canonical breakdown of her `T_mint` transfer. She would need:

- Two different `proofBytes` and/or `cms[]`, both passing the on-chain verifier.

That requires either:

- Two valid Groth16 witnesses for the same public inputs (allowed – but they yield the same `cms`, so they are not *contradictory*), or
- Two valid Groth16 witnesses with *different* public inputs. Allowed too, but each one demands its own `transferFrom` for its own `totalFace` – **that** is the duplicate-spend problem ERC-20 solves. Alice cannot transfer the same 1,200 BUCK twice without holding 2,400 BUCK to start with.

So a single 1,200-BUCK escrow yields exactly one accepted batch. The only freedom Alice has is *which* batch she lays down – which denominations and which recipients – and that freedom is hers to keep, by design.

What about hostile-Alice trying to mint *more* commitment value than she escrowed? She would need a witness whose `sum(v[i])` equals the integer `totalFace` *that the EVM sees*, but whose individual `v[i]` values *actually* sum to more. In `F_r` with $r \sim 2^{254}$ and no range check on `v[i]`, the constraint `sum(v[i]) == totalFace` would be satisfiable by witnesses whose integer sum differs from `totalFace` by a multiple of `r`. `mint_batch.circom` closes this with a `Num2Bits(128)` range constraint on each `v[i]` and on `totalFace`, mirrored by `spend.circom`’s check on `face` and the witness `v`. A 128-bit cap is comfortably above any plausible BUCK denomination and well below $r/2$, so an integer overshoot can never wrap.

What about hostile-Alice trying to mint a commitment that is *already* in the tree (to collide on a nullifier she does not control, or to attempt a hash-collision attack)? She cannot collide on a nullifier without knowing somebody else’s opening – the nullifier is a Poseidon-PRF output, so forging a collision is as hard as inverting Poseidon. She *can* re-mint her own opening (the contract no longer blocks duplicates), but doing so forces her to pay `totalFace` a second time and buys her nothing: the nullifier is deterministic in the opening, so whichever of her two copies she spends first permanently nullifies both. This is the value-conservation property in action – the chain does not need a `commitmentExists` check because the nullifier mechanism already collapses duplicate openings to one spend, and Alice bears the cost of any duplication.

7 Why Outside Observers Cannot Link

Mallory has full read access to the chain after `T_mint` confirms. What does she see?

```
T_mint:
  from: Alice (a registered IdentityRegistry account)
  to:   Notes (a system-public contract)
  value: 0 (BUCK is ERC-20, not ETH)
  data: Notes.mint(proofBytes, oldRoot, newRoot, nextLeafIndex,
                  1200e18, [cm_0..cm_15])

events emitted (in order):
  Buck.Transfer(Alice, Notes, 1200e18)
  Notes.Minted(Alice, 1200e18, startIndex=K, count=16, newRoot)
```

The batch-mint contract emits one `Minted` event per batch rather than per-leaf `Appended` events; the commitments themselves live in calldata for off-chain provers and wallets to read. (Calldata is about an order of magnitude cheaper per byte than LOG data, which is part of why the N-independent cost bound holds.)

What can Mallory infer?

- That Alice (publicly) escrowed 1,200 BUCK. **Yes** – her address is `msg.sender`.
- That sixteen new commitments entered the pool at indices `[K..K+15]`. **Yes** – `Minted` announces `startIndex` and `count`, and `cms[]` is in calldata.
- *Which* commitment is worth *how much*. **No** – each is a fresh Poseidon5 output hidden by an independent `rho_i`. Without a `rho_i`, distinguishing the 1,000-BUCK hash from the 25-BUCK hash (or from a 0-BUCK dummy) is computationally equivalent to inverting Poseidon on a uniform field element, modelled here as a random oracle.

- /Which entries are live notes and which are dummies.* **No** – the padding leaves are constructed with the same Poseidon-5 shape and a fresh **rho**; nothing about them is chain-distinguishable from a live **cm** with a small **v**.
- *Who* holds each live note. **No** – **idHash_i** is also inside the hash.
- Whether two commitments share a recipient. **No** – two notes addressed to the same **idHash** but with distinct **rho_i** produce uniformly independent **cm_i**, so the chain shows nothing distinguishable from independent draws.

What about *linking back to a specific note artifact* that Bob later possesses? Mallory sees **cm_0..cm_5** published by Alice in **T_mint**. Bob's artifact references some **cm_i** and **T_mint**, but the artifact lives off-chain. Mallory cannot get the artifact unless Bob (or Alice) shows it to her, and the chain alone gives her no way to single out **cm_i** as "the one Bob holds" against the other five.

The remaining linkability surface is the issuer. **msg.sender** is publicly Alice, so every commitment in the batch carries a public **issuer** tag. This is intentional and desired: a recipient must be able to verify who minted his note. The privacy goal is unlinkability of recipients and amounts, not of issuers.

7.1 Spend-Time Unlinkability

The shipped spend circuit (**spend.circom**; see alberta-buck-ethereum.org) lets Bob deposit his note without revealing *which* **cm** in the on-chain set he opened. The spend SNARK proves "I know an opening to *some* commitment in the Merkle tree under the public **noteRoot** that hashes to this nullifier", with **cm** as a private witness rather than a public input. Unlinkability of *spend events* to *mint events* is now a chain-observable property: Mallory sees a fresh nullifier, a public face, and a recipient – nothing that points back to a specific commitment in the tree. The full walkthrough of the redemption flow is in **The Redemption (Spend), Step by Step** below.

8 Atomicity, No Double-Mint, No Replay

Three properties together rule out duplicated or partial mints:

8.1 Atomicity

Already covered above: **Notes.mint** performs proof verification, BUCK pull, and commitment appends in one EVM call. Either all happen or none.

8.2 No Double-Mint From One Transfer

The transfer and the commitments are bound in the same call: there is no "pre-deposit, then mint" or "mint, then deposit later" flow that an attacker could desynchronize. Each **Notes.mint** invocation does its own **transferFrom**; each **transferFrom** debits Alice's BUCK balance once. Two minting transactions require two **transferFrom** debits, which require Alice to actually hold $2 * \text{totalFace BUCK}$.

8.3 No Replay

A literal replay – broadcasting the exact same `T_mint` calldata in a new transaction – fails for three independent reasons:

- a. **Stale-state guards.** The contract checks `oldRoot == roots[currentRootIndex]` and `nextLeafIndex == self.nextLeafIndex`. The first mint has already advanced both, so the replay’s `oldRoot` and `nextLeafIndex` public inputs no longer match the live chain state. The contract reverts before the SNARK is even called, with `Notes: stale oldRoot` or `Notes: stale leafIndex`.
- b. **SNARK binding.** Even if a future minter coincidentally re-created the same `oldRoot` and `nextLeafIndex` state (which is prohibited by the monotonic advancement of `nextLeafIndex`), the SNARK was bound to the specific `(cms[], newRoot)` pair of the prior batch. A replay would fold the same `cms[]` a second time and produce a different `newRoot` than the SNARK attested to, breaking the rolling-tree relation.
- c. **Allowance and balance.** Even setting the above aside, the second mint would re-debit Alice’s BUCK and re-consume her allowance to the `Notes` pool – economic-level double-escrow protection on top of the structural guards.

Replay of the same opening across batches is structurally allowed but economically self-punishing: Alice escrows `totalFace` again, the duplicate commitment lands at a fresh leaf index, and the deterministic nullifier `Poseidon3(rho, idHash, 4242)` means the first spend nullifies every copy. Alice pays twice for a spend-once outcome.

8.4 Cross-Chain or Cross-Deployment Replay

The `MintGroth16Verifier` verification key is fixed by the trusted-setup ceremony for that deployment. A proof generated against the dev-key in `scripts/snark/setup.sh` will not verify against a production-key contract on mainnet, and vice versa. Public inputs are also sealed into the proof: a proof valid on Sepolia for `totalFace=1200` cannot be reused on mainnet because the on-chain verifier address (and key) differ.

The only cross-context vulnerability would be a *chain-id collision* on identical verification keys, which the production setup ceremony precludes by construction.

9 Prover Contention and the Rollup Race

Batch minting carries one genuinely operational concern: a mint transaction is a *rollup-style* transaction. Alice’s proof is bound to the specific `oldRoot` and `nextLeafIndex` she snapshotted when she began proving; if any other mint lands between her snapshot and her submission, those values go stale and her transaction reverts at one of the stale-state guards.

This is not a *safety* concern – Alice loses no BUCK (the contract reverts before `transferFrom` runs) and produces no bad on-chain state. It is a *liveness and efficiency* concern: Alice wasted the prover time spent constructing a proof that is no longer accepted, and has to re-prove against the fresh state.

9.1 Who Wins When Two Minters Race?

When Alice and Bob both submit `T_mint_A` and `T_mint_B` targeting the same `oldRoot`:

- One of the two (by block ordering) lands first and updates the root.
- The other transaction executes, discovers the stale-state mismatch, and reverts at the `require(oldRoot == roots[currentRootIndex])` line before any BUCK moves.

Transaction fees are paid by the loser for the reverted transaction (gas for the revert path, not for the full mint). The loser then re-proves against the new `oldRoot` and `nextLeafIndex` and resubmits. The new proof folds *their* commitments onto the root that *now* includes the winner's commitments – no commitment has been lost; the loser's tree-position is just different from the one they originally predicted.

9.2 Mitigation Layers

Three mitigation layers, in ascending complexity:

1. **Optimistic retry (wallet-level).** On revert, the wallet re-fetches (`currentRoot`, `nextLeafIndex`) from the chain, rederives the sibling paths for the new insertion positions, re-runs the prover, and resubmits. For small N this is fast enough to not matter. This is the recommended initial deployment strategy.
2. **Retry budget and back-off (wallet-level).** For large N where prover time is nontrivial (minutes to hours at $N=1024$), the wallet should expose a "chance of contention" estimate to the user and offer to batch at the smallest pinned N that fits the job, or to enqueue the mint on a private relay channel during low-traffic windows.
3. **Sequencer / proof aggregation (future).** If real-world contention warrants it, an external sequencer can batch multiple users' mints into a single proof per slot, preserving the N -independent on-chain cost while amortising prover time. Sequencer capture is a separate trust axis and is deferred; the shipped protocol is sequencer-free.

9.3 What Does Not Regress

- **Atomicity.** A reverted mint leaves *no* chain-visible state change; it is strictly indistinguishable from no submission.
- **Value conservation.** No reverted transaction debits BUCK, so there is no way for a loser to pay the pool without appending commitments.
- **Unlinkability.** A re-submitted mint on a fresh root still produces the same (anonymity-preserving) calldata and event shape; Mallory cannot tell a retry from a first attempt.
- **Spend-time invariants.** The spend circuit takes the current `noteRoot` (or any recent-window root); a late-arriving mint does not invalidate any previously-proved spend.

The rollup race is the price of replacing $\sim 800K$ gas/leaf with ~ 500 gas/leaf, and for practical block-rate contention it is negligible.

10 Receipt Construction From the Transaction ID Alone

A practical question: months later, Bob wants to prove to Carol that his note was legitimately issued. He needs no help from Alice and no live indexer:

1. Bob fetches `T_mint` from any Ethereum archive node.
2. He decodes the call data into (`proofBytes`, `oldRoot`, `newRoot`, `nextLeafIndex`, `totalFace`, `cms`) using the standard `Notes.mint` ABI.

3. He notes `msg.sender` and resolves it to Alice's identity via the IdentityRegistry on the same chain (a single `eth_call`).
4. He locates his `cm` in `cms[]` at `leafIndex - nextLeafIndex` and asserts equality.
5. He recomputes `Poseidon5(flavor, v, rho, idHash, predicate) == cm` from the opening.
6. (Optional.) He re-runs `MintGroth16Verifier.verify(proofBytes, [issuerMode[0..N], oldRoot, newRoot, nextLeafIndex, totalFace, cm[0..N]])` against the per-N verifier keyed by `cms.length`, assembling the public-signal vector from the calldata exactly as the contract does, convincing himself the EVM did its job.

The "receipt" Carol receives is therefore `(T_mint, opening, idHash mapping)`. Its validity is independently checkable against the public chain. Nothing here requires ongoing cooperation from Alice – which is exactly the property a receipt should have.

11 The Redemption (Spend), Step by Step

This section walks the *inverse* of the mint. Bob holds the opening for the 1,000-BUCK note `cm_0`; he wants to redeem it for 1,000 BUCK credited to his on-chain account, without revealing *which* commitment in the pool he consumed.

There is **one** spend path, the *Identity-M-bound deposit*: Bob redeems by proving he is the KYC-registered Identity the note names – *not* a particular account – so he may use *any* Fountain account bound to his Identity point M_{rec} , and he reveals no Identity to the chain. Every spend composes the same two halves over a single committed point P :

1. **The note proof** (this section, Steps 1–3): the kept, flavor-agnostic `spend.circom` Groth16 proves the note's commitment `cm` is in the pool tree under a public `noteRoot` and emits a fresh, well-formed `nullifier` – hiding *which* commitment was consumed. `Notes.spendCoupled{A1,A2,B1}` all reuse `spendVerifier.verifySpend` for exactly this.
2. **The identity binding** (Step 4): an EIP-196 Okamoto sigma + a membership proof bound to the *same* P , proving the spender is a registered Identity. The flavor selects which sigma and what P commits – the issuer's M_I (A2), or the spender's own $M_{\text{rec}}/M_{\text{dep}}$ (A1/B1). This is detailed per-flavor in 12; the walk below uses Bob's **A2** note (addressed, private issuer) as the running example.

Three properties of the flow are worth stating up front, because they shape every step:

- **The anonymity set is the full pool, not only same-denomination.** The spend proof takes `cm` as a *private* witness under a public `noteRoot` – the chain-observable anonymity set is every unspent commitment in the tree. The **face value is a public output**, because the BUCK contract must know how much to pay out; amount-level linkability narrows the effective set to commitments whose hidden `v` could equal the revealed **face**. Same-denomination issuance is therefore the strongest-privacy regime, but the underlying cryptographic hiding is over the whole tree, not over a denomination class.
- **The BUCK ERC-20 does not verify the SNARK itself.** The authorization chain is: valid SNARK + accepted root + fresh nullifier (+ for A-spends, valid identity binding) → the relevant `spendCoupled{A1,A2,B1}` entrypoint proceeds → the entrypoint invokes

`buck.transfer(recipient, face)` from the pool (Buck routes it through the Carrying path). The proof is a capability token for entry into the appropriate spend method; the resulting ERC-20 movement is a routine pool-to-recipient call gated by that entry. Since the `Notes` pool is a system-public account in the IdentityRegistry, the pool’s outbound transfer does not itself require a Chaum-Pedersen receipt – the recipient’s own registration is what makes them acceptable as a transfer destination.

- **Public inputs are bound into the proof, including `chainId`.** `spend.circom` ghost-binds `recipient` and `chainId` (via `x*x` rows) so Groth16’s IC[] commitment makes them non-malleable from the mempool. A front-runner who steals Bob’s `proofBytes` and resubmits with a different `recipient` or on a different chain id is rejected by the verifier.

11.1 Step 0: Bob Holds a Verified Note

From the mint walk (**Step 5: Recipient Verification**) Bob already has:

```
note_artifact_0 = {
  txHash:    T_mint,
  index:     K + 0,
  cm:        cm_0,
  opening:   ( flavor_A1, 1000e18, rho_0, idHash_bob, predicate_0 ),
  meta:      { issuer: Alice, ... },
}
```

He has independently confirmed that `cm_0` is in `Notes.commitments[]` at index `K` and that `Poseidon5(opening)` reproduces it. The opening has been sitting in his wallet for weeks, months, or years – addressed A-notes are cryptographically safe for indefinite holding (`Notes`).

11.2 Step 1: Fetch the Commitment Tree and Build a Merkle Path

`Notes.sol` maintains a Tornado-style incremental Poseidon-T3 Merkle tree of `TREE_DEPTH = 20` (max ~1M leaves). The per-leaf insertion math lives inside the mint SNARK; the contract just records the SNARK-attested `newRoot` at each `mint()` call. The current root is exposed as `noteRoot()`. Two things happen:

- Bob (or, more practically, any light-node helper running off-chain) reconstructs the tree off-chain. The commitments themselves are available from each mint transaction’s calldata `cms[]`, keyed by the `Minted(issuer, totalFace, startIndex, count, newRoot)` event that indexes the batch. A replayer walks all `Minted` events in order, takes the `cms[]` slice out of each mint’s calldata, and appends them to its local tree using the same `ZERO_VALUE` and Poseidon-T3 wrapper the in-circuit insertion uses. Bob’s Merkle path for `cm_0` – a sequence of sibling hashes plus left/right indices up to the root – falls out of that rebuilt tree. The path is purely public information; everyone can reproduce it, and Bob does not have to trust any indexer because calldata is canonical.
- Bob reads the current `noteRoot()` via `eth_call`. He may instead pin any root in the recent-roots window: `Notes.sol` retains the last `ROOT_HISTORY_SIZE = 30` roots in a ring buffer (`roots[]`, advanced once per inserted leaf), and the on-chain `isAcceptedRoot()` scans that ring backward from `currentRootIndex`. This is the standard zk-rollup hygiene (Zcash, Tornado Cash) for tolerating prover latency: a spend assembled against a slightly-stale root remains acceptable so long as fewer than 30 mints have landed in the meantime.

11.3 Step 2: Compute the Nullifier

The shipped `spend.circom` derives the nullifier as a Poseidon-3 hash of the witness `rho`, the witness `idHash` bound at mint, and a constant tag:

$$nf = \text{Poseidon}_3(\rho_0, \text{idHash}_0, 4242).$$

The tag (4242) is a fixed nonzero field constant that domain-separates nullifier preimages from the Poseidon-5 commitment preimages, so a structural collision between a nullifier and a commitment is impossible. Three properties follow:

- **Deterministic.** The same `(rho, idHash)` always yields the same `nf`. So Bob can only spend `cm_0` once; a second attempt re-derives the same `nf` and is rejected by `nullifiers[nf] = true` on the first spend.
- **Unforgeable without the opening.** An attacker who sees `cm_0` and the tree cannot compute `nf` without `rho_0` and `idHash_0`, both of which live inside the opening (and `rho_0` is not derivable from the recipient's identity alone).
- **The note proof binds the opening and nullifier, not the identity.** The flavor-agnostic `spend.circom` treats `idHash` as a witness scalar – it does not constrain it to a registered Identity. That binding is the *separate* Step-4 deposit gate: for A1/A2 one folded proof that recomputes this same nullifier from the *idHash* it opens, and shows the spender is the registered Identity paired with the key that reads the note; for B1 the EIP-196 depositor-binding sigma and a membership proof over its committed point. Keeping the identity binding out of `spend.circom` keeps the one proof every spend shares small. And because the deposit is *identity*-keyed (not account-keyed), an addressed note survives the loss of any single account key: re-register a fresh Fountain account on the same M_{rec} and deposit (recovery, not finality).

The nullifier is the chain-observable anchor of the spend. Publishing it marks `cm_0` as consumed from the tree's *semantic* point of view, even though the tree itself remains append-only and nothing is deleted.

11.4 Step 3: Generate the Spend Proof

Bob's wallet runs `scripts/snark/prove_spend.js` (the spend-side analogue of `prove_mint_batch.js`, using the depth-20 PTAU produced by `scripts/snark/setup.sh`). `spend.circom`'s signal layout is exactly:

```
public noteRoot;           # an accepted root from the on-chain ring buffer
public nullifier;          # the nf from Step 2
public face;               # 1000e18 -- the BUCK to release (range-bounded to 128 bits)
public recipient;          # Bob's on-chain address (the payout destination)
public chainId;            # replay-domain separation

private flavor, v, rho,
      idHash, predicate;    # the Poseidon-5 opening
private pathElements[depth]; # depth-20 sibling hashes
private pathIndices[depth];  # left/right bit per level
```

The shipped circuit enforces five relation groups:

1. **Range bounds and value match.** Both the public `face` and the witness `v` pass through `Num2Bits(128)`, then `face === v`. This closes the field-overflow gap: `v` cannot wrap around `r` to claim more than its in-range share.

2. **Opening binds.** `cm = Poseidon5(flavor, v, rho, idHash, predicate)`.
3. **Merkle membership.** `MerkleProof(20)` walks (`cm`, `pathElements`, `pathIndices`) up the tree using `Switcher + Poseidon(2)` per level and asserts the climb reaches `noteRoot`.
4. **Nullifier derivation.** `nullifier === Poseidon3(rho, idHash, 4242)`.
5. **Public-input ghost binding.** `recipient` and `chainId` each appear in a trivial $x \times x$ row. This is the minimal anchor needed for circom not to optimise out unreferenced public signals; once anchored, Groth16's IC $\|$ commitment makes them non-malleable from the mempool's perspective.

Critically, `cm` is *not* a public input; it is recomputed inside the proof from the witness opening. The proof convinces the verifier that *some* `cm` in the tree satisfies (1)-(4); it does not reveal *which*.

This `spend.circom` note proof is *flavor-agnostic*: A1, A2, and B1 all use the same circuit and the same on-chain `spendVerifier.verifySpend` check for the `cm`-membership + nullifier discharge (nullifier tag 4242). Bob's wallet serializes the Groth16 proof into the standard (`uint256[2]`, `uint256[2][2]`, `uint256[2]`) ABI tuple (256 bytes). The flavor-specific *identity* material – the re-encryption `eEnc` and the fold proof (A1/A2), or `eDepForIss`, the depositor-binding proof and its membership proof (B1) – is assembled separately and submitted alongside the note proof in Step 4.

11.5 Step 4: Submit `Notes.spendCoupledA2` On Chain

Bob's note is A2 (private issuer), so he deposits via `spendCoupledA2`. Beyond the Step 1–3 note proof he supplies exactly two things: `eEnc` – a *fresh re-encryption* of the leaf ciphertext `eIss` under his mailbox key pk_{recv} (same plaintext, new randomness, so the chain never sees the mint-time ciphertext again) – and `foldProof`, the folded gate's Groth16 triple (`alberta_buck.wallet.deposit_fold`):

```
notes.spendCoupledA2(proofBytes, noteRoot, identityRoot, nullifier, 1000e18, bob,
                    eEnc, foldProof);
```

That is the whole addressed calldata, with the identity root Bob proved against. A sigma, a separate membership proof and a note-binding proof used to ride along beside it, sharing a public point; one witness states what they only jointly suggested, so they are gone and so is the point. The honest EVM walks:

```
require(address(identityRegistry) != address(0), "Notes: identity registry not set");
require(recipient != address(0), "Notes: zero recipient");
require(face > 0, "Notes: zero face");
require(!_isAcceptedRoot(root), "Notes: unknown root");
require(!nullifiers>nullifier], "Notes: already spent");

// (1) Note proof: cm in the pool tree under 'root', nullifier well-formed,
// flavor bound to this entry point.
require(spendVerifier.verifySpend(
    proof, root, nullifier, face, recipient, block.chainid, FLAVOR_A2, 0
),
    "Notes: bad spend proof");

// (2) The identity root: one the registry still retains, posted within
// the age the Notes consumer declares (seven days).
require(identityRegistry.acceptsRoot(identityRoot, NOTES_MEMBERSHIP_CONSUMER),
    "Notes: identity root not accepted");

// (3) The folded gate. The adapter reads msg.sender's registered key and
// credential and derives EVERY public input, so relation (2) is
// necessarily about an account that really is registered.
require(depositFoldVerifier.verifyFoldA2(
```

```

    foldProof, nullifier, identityRoot, eEnc, msg.sender),
        "Notes: bad folded deposit gate");

nullifiers[nullifier] = true;
noteFaceSum          -= face;

require(buck.transfer(recipient, face),    // pool isCarrying => Carrying path
        "Notes: transfer failed");
emit SpentCoupledA2(nullifier, face, recipient);

```

Five invariants are enforced in order:

a. **root** appears in the recent-roots window. A root that has rotated out is stale – Bob must rebuild the proof against a fresher root. b. **nullifier** has never been published before (Theorem 10 in the proofs says this single-bit check is sufficient for double-spend prevention). c. The note proof is valid against `SpendGroth16Verifier` via the `SpendVerifierAdapter`, which packs `[root, nullifier, face, recipient, chainid]` into the public-signal vector. By soundness Bob holds a Poseidon-5 opening of a `cm` under `root`, with `nullifier = Poseidon3(rho, idHash, 4242)`. d. `identityRoot` is accepted: the registry still holds it in its ring of 256 postings, and it was posted within the seven days `Notes` declares. That age is how long a holder cleared from its registry can still spend. e. `verifyFoldA2` succeeds. One proof shows that k opens `eEnc`, that `msg.sender`'s registered credential opens to the same m_{rec} the receiving leaf commits beside k , that the decrypted issuer Identity is registered under the shipped salt, that the note's committed T opens against k , and that the nullifier is the one this note's `idHash` produces. The adapter derives the public inputs, including the account read from the registry. There is no committed point in the statement, and the event logs none.

If any of (a)-(e) reverts, the whole transaction reverts: no BUCK moves, the nullifier bit is never set, `noteFaceSum` is unchanged. **Bob gets one attempt per root window**; a failed spend retries with a refreshed proof, but never consumes a nullifier without paying out the BUCK.

An A1 note takes `spendCoupledA1`, the same fold with the face public and no issuer-membership relation: `eRec` encrypts the plaintext M_{rec} under pk_{recv} . A B1 note takes `spendCoupledB1`, where the sigma is `verifyDepositorBinding` and the membership proof opens P_{dep} on H_{PEDERSEN} . All three are detailed in 12.

11.6 Step 5: BUCK ERC-20 Pays Out (the Carrying path)

The call inside `Notes.spendCoupledA2` is a plain `buck.transfer(bob, 1000e18)`. Because the `Notes` pool is bound `isCarrying` in the `IdentityRegistry`, BUCK routes that transfer through its internal `_carryingTransfer` path on this single egress. (`spendCoupledA1` and `spendCoupledB1` take the same path.) Nothing in `Buck.sol` is special-cased for the pool: it is just a registered, `isCarrying` account making an outbound transfer of 1,000 BUCK to a registered recipient. The result on-chain:

- `Buck.balanceOf(Notes)` decreases by exactly 1,000 BUCK – nominal balance only.
- `Buck.balanceOf(Bob)` increases by exactly 1,000 BUCK.
- `Buck.Transfer(Notes, Bob, 1000e18)` is emitted (a standard ERC-20 event; the carrying apportionment is internal, with no distinct event).
- `Notes.Spent(nullifier, 1000e18, Bob)` is emitted.
- `Notes.nullifiers[nullifier]` becomes `true`.

What "carrying" means in the ledger. BUCK demurrage is a balance-time integral: each account stores its `buckSeconds` (the crystallised $\int B dt$) plus the `timestamp` of its last touch. A transfer *from a non-Carrying account* settles the sender's accrued fee (a one-shot burn) and hands the recipient fresh, zero-age BUCK. A transfer from an `isCarrying` account – the `_carryingTransfer` path – differs in two surgical ways:

1. **No sender-side fee burn.** The pool sheds a proportional slice of its live buck-seconds rather than burning a fee; its per-BUCK age is preserved.
2. **Recipient absorbs the pool's age.** The recipient inherits that slice of buck-seconds (`carried = liveBs * amount / poolRaw`), added to its own:

```
liveBs    = pool.buckSeconds + pool.raw * elapsed;
carried   = liveBs * amount / pool.raw;
pool.buckSeconds = liveBs - carried;    // pool sheds its share
recipient.buckSeconds += carried;      // recipient inherits the age
```

System-wide BUCK-age is preserved across the call (no fee is burned, no fresh-BUCK dilution at the recipient); the demurrage that the pool would have burned on a standard outgoing transfer instead rides into Bob's account as inherited buck-seconds. Bob will pay it the next time he makes a standard outgoing transfer.

Why this is the right rule for the pool. Per-note age is opaque – the pool holds aggregate BUCK behind opaque commitments and cannot tell Bob's note's age from any other note's. The Carrying path collapses the calculation to the pool's average per-BUCK age at spend time: every spender of the same `face` absorbs the same buck-seconds slice, keeping the spend-time anonymity set intact. The trade-off is that Bob inherits the pool's average age regardless of how long *his specific* note sat in the tree. Anyone whose note rested longer than the pool average comes out ahead; this is the cost of anonymity, and the upside for long-term holders.

The full discussion (one new BUCK method, one new event, no per-account mode flag) is in **Demurrage-Carrying Transfers** below.

11.7 What the Observer Sees at Spend (Mallory Returns)

After `T_spend` confirms, Mallory can read:

```
T_spend:
  from: Bob (a registered IdentityRegistry account)
  to:   Notes (system-public)
  data: Notes.spendCoupledA2(proofBytes, noteRoot, identityRoot, nullifier,
                             1000e18, Bob, eEnc, foldProof)

events:
  Buck.Transfer(Notes, Bob, 1000e18)
  Notes.SpentCoupledA2(nullifier, 1000e18, Bob)
```

What she infers:

- **That a note was spent.** Yes – the nullifier is fresh and the pool balance dropped by 1,000 BUCK.
- **That the spend was for 1,000 BUCK to Bob.** Yes – `face` and `recipient` are public inputs (they must be, to tell the ERC-20 whom to pay).

- *Which commitment in the tree was consumed?* **No** – `cm` is a private witness to the SNARK. Mallory’s anonymity set is every unspent commitment of flavor `A` in the pool whose hidden `v` could equal 1,000 BUCK. At deployment scale this is potentially the entire pool minus prior nullified entries; in a same-denomination regime (many 1,000-BUCK notes in circulation) it is exactly the 1,000-BUCK subset.
- *Which mint did the spent note come from?* **No** – there is no cross-reference to `T_mint` in `T_spend`. The only link back to any specific mint would be the nullifier, but nullifiers are Poseidon-PRF outputs over secret material; inverting one to recover (`flavor`, `M_rec`, `rho`) is exactly the pre-image resistance we assume.
- *Who held the note before Bob?* **No**. If Alice gave the note to Carol who handed it to Dave who sold it to Bob, none of those transfers touched the chain. The chain sees only "at mint Alice deposited, later Bob withdrew" – the intermediate holders are invisible.

The residual linkability surfaces are (i) **face** (the revealed amount) and (ii) **recipient** (the on-chain payout address). Both are unavoidable at the SNARK/ERC-20 boundary – the contract must know where and how much to pay. The community- standardized mitigation is fixed-denomination notes (so revealing **face** does not subdivide the anonymity set) and batched, relay-mediated payouts (so revealing **recipient** need not reveal the *true* beneficiary). Both are orthogonal to the core spend circuit and can be layered on top.

11.8 Why Alice Cannot Steal Bob’s Note (Even Between Rooms)

Can *any* actor other than Bob – Alice-the-issuer, a prior holder, or a random Mallory – redeem `cm_0`? Three protections apply, the third being the folded deposit gate:

1. **Opening secrecy.** `spend.circum` requires the full Poseidon-5 opening (`flavor`, `v`, `rho`, `idHash`, `predicate`) as a private witness. Without `rho_0` and `idHash_0` the circuit is unsatisfiable, so no one without the opening can produce a valid nullifier or proof. For Alice’s secrets, this is the load-bearing protection on bearer-flavor (B1) notes: she could in principle race the bearer to deposit, which is the civil-trust semantics of bearer cash and is discussed at length in alberta-buck-notes.org §Long-Term Holding Safety.
2. **Recipient-binding at the payout site.** `recipient` is a public SNARK input ghost-bound by `recipient * recipient`, so Groth16’s IC $\parallel$ commitment makes it non-malleable. A front-runner who intercepts Bob’s `proofBytes` and resubmits with a different `recipient` is rejected by the verifier. `chainId` is bound identically, blocking cross-chain replay.
3. **Identity binding via the folded gate.** For an A2 deposit Bob provides the ciphertext and one Groth16 proof. With the fold verifier set, `spendCoupledA2` checks that the receiving secret opens the ciphertext, that Bob’s registered account opens to the Identity paired with that secret in the receiving leaf, and that the decrypted issuer is registered under the shipped salt. Alice cannot redeem to herself even knowing the opening: she does not hold `k`, and no leaf pairs her Identity with Bob’s mailbox.

The strict identity-binding guarantee is a chain-observable property when the fold verifier is set: one Groth16 accept is the gate. The note proof enforces the commitment and the nullifier; the fold enforces that the depositor is the Identity paired with the mailbox the note was keyed to.

11.9 B1 Bearer Notes

A B1 note (bearer, public issuer) is redeemed the same way, with the depositor binding playing the role the fold does for A2: `spendCoupledB1` calls `verifyDepositorBinding` (the depositor re-encrypts its own M_{dep} under the public issuer’s key, coupled to its payout account) plus a membership proof of M_{dep} . Knowledge of the Poseidon-5 opening is still the *bearer* authorisation – if two parties hold copies of the same B1 opening (a paper note photographed and forwarded), whichever submits first wins, the intended civil-trust semantics of bearer cash – but the depositor is additionally proven to be a registered Identity the issuer can name (see 12).

11.10 Recapping Property C (Only Valid Notes Redeem)

The original Property C question – *subsequently only those N valid notes can be deposited* – becomes concrete at this layer. Claim: no actor can redeem a commitment `cm` unless

a. `cm` was appended to `commitments[]` by a prior `Notes.mint` transaction whose Groth16 proof verified (i.e., `cm` is part of an honest mint batch), and b. The actor holds the opening of `cm`, and c. For A-spends, the actor is the depositor keyed by `M_rec` in that opening.

Proof sketch. The spend circuit’s relation 2 (Merkle membership) rejects any `cm` not in `noteRoot`. The mint circuit’s relations 1-2 (opening binding and sum conservation) are what places a `cm` in `noteRoot` in the first place, and by the mint-side soundness story in **Why Alice Cannot Cheat** every such `cm` corresponds to a well-formed opening whose `v` is included in the mint’s `totalFace` escrow. The spend circuit’s relation 5 (`face == v`) then ensures the BUCK released equals exactly the share of the escrow that `cm` was allocated. Summing over every accepted spend of a given mint batch, the cumulative BUCK released equals at most the `totalFace` escrowed – property C follows from mint-side value conservation plus spend-side nullifier uniqueness.

Field wrap-around is foreclosed by range bounds: `mint_batch.circom` range-bounds each `v[i]` to 128 bits and `spend.circom` range-bounds both `face` and the witness `v` identically, so an integer overshoot of `totalFace` by a multiple of `r` is structurally impossible at either end. Property C rests on those range bounds plus the nullifier-uniqueness check and Merkle membership.

12 The Identity-M Spend Path (A1, A2, B1)

Step 4 above showed the A2 deposit in the redemption walk; this section gives the per-flavour detail for all three (the notes-identity-axis unification). The three entry points are not one gadget. Addressed spends are the folded proof. The bearer spend is a sigma beside a membership proof, because its two claims share a secret and the addressed claims do not. Three entrypoints:

- **A2 (addressed, private issuer)** – `Notes.spendCoupledA2` calls `verifyFoldA2`. The ciphertext is keyed to pk_{recv} . The proof shows k opens it to a registered issuer Identity, under the salt the issuer shipped, and that the depositor’s account is the Identity paired with that k in the receiving leaf.
- **A1 (addressed, public issuer)** – `Notes.spendCoupledA1` calls `verifyFoldA1`. `eRec` encrypts the plaintext M_{rec} under pk_{recv} . The issuer is public, named at mint by the batch Schnorr. The face is a public input of the fold.
- **B1 (bearer, public issuer)** – `Notes.spendCoupledB1`. The depositor encrypts its own M_{dep} under the issuer’s registered account key and proves `verifyDepositorBinding`. The membership proof is `identity_membership_b1`, bound to the sigma’s P_{dep} by the contract passing that point in. The issuer decrypts the event.

Each spend names the identity root it proved against, and `Notes` accepts one the registry retains and posted within seven days. Roots are posted by the aggregator the registry’s root authority appoints. A receiving leaf is admitted by its registry’s certification, since `register` refuses a caller-supplied leaf and must: a registration that accepted any leaf would accept a leaf pairing any Identity with any key, which is precisely what relation (3) relies on being impossible. Addressed membership is inside the fold, as $\text{Poseidon}(\tau_R, m, k, \text{salt})$. B1’s membership is a separate proof of a salted coordinate leaf. Both prove 32-level paths: the registry’s subtree, then the aggregator.

The note tie for A1 and A2 is inside the fold: the circuit recomputes the nullifier from the `idHash` it opens, so a substituted ciphertext is a different note. Nothing beside the fold verifies an addressed spend.

12.1 The Shared Primitives

Before the flavour-specific walkthroughs, three primitives are common to all three:

- **The identity accumulator.** Registries keep private subtrees of depth 12, an aggregator of depth 20 has their roots for leaves, and its root is posted on chain into a ring of 256, each with its time. `register` refuses a caller-supplied leaf. Every leaf function leads with its own v2 field tag τ : the addressed gate proves $\text{Poseidon}(\tau_R, m, k, \text{salt})$, and B1 proves $\text{Poseidon}(\tau_S, M.x, M.y, \text{salt})$. The unsalted point leaf remains for public trees and for the unilateral receipt’s `contains` check, which is a different statement from the spend.
- **B1’s membership proof.** `identity_membership_b1.circom` proves $T = bH_{\text{PEDERSEN}}$, $P_{\text{dep}} = M + T$ with the addition’s precondition enforced, and the salted leaf’s path. The contract passes the sigma’s P_{dep} into the verifier. The generator is hashed to the curve, so the blind cannot be shifted onto another registered Identity – a circuit that WITNESSED T instead, on a generator whose logarithm is known, let an unregistered depositor spend.

- **The aggregator.** Registries, feature authorities and regulators feed subtree roots into a `CentralMerkleService` (depth 20, up to 2^{20} subtrees) whose root the registry’s ring retains. Every level hashes alike, so a subtree path and the aggregator path compose into the one 32-level path the gates prove. The reference is `alberta_buck.registry.merkle_service`; Rust and JavaScript mirror it.

12.2 A2 Identity-Targeted Spend (The Unilateral Receipt Flow)

Bob’s note is flavour A2 (addressed, private issuer). The note opening carries $eIss = (R_e, C_e)$ – the issuer’s Identity M_I encrypted under Bob’s receiving key pk_{recv} , plus the anti-framing binding (`verifyIssuerReenc`) that ties $eIss$ to the issuer’s own registered Identity, plus the issuer’s naming salt and the randomness r' . Bob decrypts with k , not with m_{rec} .

==== Step A0: Bob Holds the A2 Note ====

The artifact Bob received off chain:

```
note = (
  flavor = A2,  v = 1000e18,  rho,  idHash,  predicate,  # the commitment opening
  eIss = (R_e, C_e),                                     # M_I under pk_recv
  T,                                                    # committed in idHash
  r_prime, gamma, salt_iss,                             # what the fold needs
  binding,                                             # verifyIssuerReenc
  mint_tx_ref,                                         # where cm was folded
)
```

==== Step A1: Bob Chooses a Deposit Account ====

Bob selects *any* registered Fountain account bound to his identity scalar m_{rec} . His wallet reads the account’s registered $(pk_{dep}, E_{dep} = (R_d, C_d))$ from the IdentityRegistry. Because all accounts share the same m_{rec} , he is free to use a fresh account every time – no re-registration, no key pinning.

==== Step A2: Build the Folded Witness ====

Bob builds ONE witness (`alberta_buck.wallet.deposit_fold`), and the gate is one proof over it. Two facts have to be established and they rest on two different secrets – k opens the note, m_{rec} owns the account – so there is no scalar for a sigma’s shared nonce to tie, and proving them side by side would say nothing about their owner. A thief with a stolen payload answers the reading half with the stolen k and the Identity half with its OWN registered Identity, and both halves are true.

```
private: k, m_rec, sk_dep, salt, r', gamma, salt_iss, two Merkle paths,
         eNote, eIss, T
public:  nullifier, identityRoot, eEnc, and -- read from the registry, not
         accepted from the prover -- pk_dep and E_dep for msg.sender
```

The relations, all in `circuits/deposit_fold_a2.circom`:

1. k decrypts the note’s ciphertext to a point M_I , and $eEnc$ re-encrypts that same M_I under the same k .
2. The account credential decrypts under sk_{dep} to $M_{rec} = m_{rec}G$.
3. $\text{Poseidon}(\tau_R, m_{rec}, k, salt)$ is a registered leaf. This is the relation a split gate leaves out, and the one the thief cannot satisfy: no registered leaf pairs ITS Identity with the key it stole.
4. That leaf’s path folds to the posted `identityRoot`.
5. M_I is itself registered, as `identity_leaf_salted( $M_I$ ,  $salt_{iss}$ )` under the salt the note shipped.

6. The note's committed T opens as $(r'k)G + \gamma H_{\text{PEDERSEN}}$: the key the mint binding proved about is the key Bob holds.

Relation (3) used to be free, because the two scalars were one value. It is now the load-bearing one. And because the circuit hashes the very signals its other relations consume, a single tampered m_{rec} breaks the decryption, the credential and the leaf at once – there is no seam between the two secrets for a thief to work in.

A1 is the same gate with relations (5) and (6) removed and the face made public: its ciphertext decrypts to the recipient's OWN Identity, so relation (2) already covers what (5) would say, its public issuer has no key to re-aim, and it ships no witness material about anyone else.

==== Step A3: What the Prover Does Not Choose ====

`DepositFoldVerifierAdapter` derives every public input on chain. It reads pk_{dep} and E_{dep} from `IdentityRegistry` for `msg.sender`, takes `identityRoot` from the registry, and limb-encodes `eEnc` itself. So relation (2) is necessarily about an account that really is registered, and there is no committed point for the wallet to decouple from anything – because there is no second proof to decouple it from.

==== Step A4: Submit the Deposit ====

```
notes.spendCoupledA2(
    proof,                // Groth16: note opening + noteRoot membership
    noteRoot,
    identityRoot,          // the posted root the fold proves against
    nullifier,             // Poseidon(rho, idHash, 4242)
    face,                 // 1000e18
    recipient,            // Bob's payout address
    eIss,                 // eEnc: the fresh re-encryption under pk_recv
    foldProof,            // the folded gate's (a, b, c)
);
```

==== Step A5: On-Chain Verification ====

`spendCoupledA2` gates on five conditions:

1. The fold verifier is wired. It cannot be cleared, so this is a deployment invariant rather than a configuration; an addressed spend without it reverts here, before anything else is checked.
2. Standard note spend: `noteRoot` accepted, `nullifier` fresh, the Poseidon-5 opening proof verifies (`spendVerifier.verifySpend`).
3. `identityRoot` is accepted for the Notes consumer: retained in the registry's ring and posted within seven days.
4. The folded gate: `verifyFoldA2(foldProof, nullifier, identityRoot, eEnc, msg.sender)` accepts.
5. BUCK transfer, then `nullifiers>nullifier` = true and `SpentCoupledA2(nullifier, face, recipient)`.

The event publishes no point. An earlier shape logged the committed P_I ; a log is not a constraint, and logging a point that no relation reads invites an auditor to believe otherwise.

If any check fails the whole transaction reverts. No BUCK moves, no nullifier is consumed. (End to end, with real proofs: `NotesE2E.t.sol`.)

What relation (5) closes. Suppose an issuer keys `eIss` to a throwaway point so the recipient cannot name it. Relations (1) to (4) still pass – Bob really can decrypt to *some* point. But that

point is not a registered Identity, so no salt opens a leaf for it and relation (5) has no witness: **the deposit reverts**. There is no spendable-but-unnameable note.

What relation (6) closes. An issuer can instead key the ciphertext so that the key its binding proves about opens it to its own M_A while Bob’s key opens it to a DIFFERENT registered Identity M_B . The binding is true and relation (5) holds on M_B ’s leaf, so without (6) Bob would honestly name M_B . But the note commits the binding’s $T = r'pk + \gamma H_{\text{PEDERSEN}}$, and (6) opens it against Bob’s own k : re-aiming the key needs $M_B - M_A$ as a known multiple of H_{PEDERSEN} , a discrete logarithm (*Proofs*, Theorem 12). The executed transcript below mints exactly this note and watches the delivery, the fold and the receipt each refuse it.

==== Step A6: The Unilateral Receipt ====

After the spend confirms, Bob’s wallet produces a *unilateral receipt* naming both parties – from k and his own Identity, two values rather than one derived from the other:

$$\begin{aligned} M_I &= C_e - kR_e && \text{(the issuer Identity);} \\ M_{\text{rec}} &= m_{\text{rec}}G && \text{(the recipient’s own).} \end{aligned}$$

A *verifiable decryption* proof (`verifiable_decrypt_prove`, a Chaum-Pedersen DLEQ) makes the receipt third-party-checkable without revealing k . The receipt tuple:

$$\mathcal{R} = (M_I, M_{\text{rec}}, pk_{\text{recv}}, v, eIss, \text{vd}, \text{binding}, \gamma, \text{issuer addr}, \text{chainid}).$$

A court or auditor checks five links (`verify_receipt`):

#	Check	Establishes
1	<code>verifyIssuerReenc</code> vs. issuer’s on-chain record	$eIss$ carries the issuer’s <i>registered</i> M_I (no framing)
2	<code>verifiable_decrypt_verify(eIss, pk_recv, M_I)</code>	$eIss$ decrypts under pk_{recv} to M_I
3	$C_e - T + \gamma H_{\text{PEDERSEN}} = M_I$	the binding’s Identity is the one decrypted (the key
4	$\ell(M_I) \in \text{rt}_{\text{id}}$	the issuer Identity is registered
5	$\ell(M_{\text{rec}}) \in \text{rt}_{\text{id}}$	the recipient Identity is registered

Full specification and tests: *alberta-buck-notes-unilateral*.

12.3 A2 in Code: The Full Cycle, End to End

The cryptography above is small enough to run. This section is a complete, *executing* transcript against the shipped wallet (`alberta_buck.wallet.unilateral_a2` over BN254 $\mathbb{G}_1$) – mirroring `test_unilateral_a2.py`. The blocks below are org-babel source blocks sharing one Python session: re-exporting this document re-runs them top to bottom, and the printed **RESULTS** are regenerated. The session is seeded, so a re-run reproduces this transcript byte for byte – and fails loudly if the wallet API drifts. The wallet’s curve arithmetic executes on the compiled `buck-identity` kernel (`buck_core.buck_identity`, arkworks BN254); the first block asserts it, so this document renders only against the production cryptography. The pure-Python `py_ecc` path remains the executable specification the kernel is proven bit-identical against (`core/vectors/identity-kernel-vectors.json`).

The cast: two Identities, one of them with two accounts. An *Identity* is a KYC-bound scalar m and its point $M = mG$. The Identity Fountain lets one m back unlimited unlinkable *accounts*, each an ElGamal encryption $E_{\text{addr}} = (R, C) = (rG, M + rpk)$ of the *same* M under that account’s own key.

```

import random
from alberta_buck.wallet import backend
from alberta_buck.wallet.bn254 import G1, ORDER, mul, add, neg, eq, rand_scalar
from alberta_buck.wallet.elgamal import elgamal_encrypt, elgamal_decrypt
from alberta_buck.wallet.unilateral_a2 import (
    IdentityTree, mint_unilateral_a2,
    make_receipt, verify_receipt,
)
from alberta_buck.wallet.recvkey import receiving_key
from alberta_buck.wallet.salt import derive_salt
from alberta_buck.wallet.deposit_fold import (
    DepositFoldRefused, deposit_fold_check, deposit_fold_a2_witness,
    deposit_fold_witness,
)
from alberta_buck.registry.tree import identity_leaf_salted, receiving_leaf
from alberta_buck.registry.merkle_service import rooted_registry

assert backend() == "kernel", "must run the production kernel: evaluate in the repo venv"
print(f"wallet crypto backend = {backend()} (buck_core.buck_identity)")

_seed = random.Random(0xA1BE27AB0CC)      # ONE deterministic transcript
rng = lambda: _seed.getrandbits(256)

CHAINID = 1
def account(m):
    # one Fountain account bound to identity m
    sk = rand_scalar(rng); pk = mul(G1, sk)
    M = mul(G1, m)                # the shared Identity point
    r = rand_scalar(rng)          # registration randomness; the fold witnesses it
    E = elgamal_encrypt(M, pk, r)  # registered credential
    return dict(m=m, M=M, sk=sk, pk=pk, E=E, r=r)

def pt(P):
    # compact point display
    return f"({int(P[0]) % 10**8:>8d}..., {int(P[1]) % 10**8:>8d}...)"

m_iss, m_rec = rand_scalar(rng), rand_scalar(rng)
issuer = account(m_iss)           # the payer (private issuer)
rec0 = account(m_rec)            # recipient account #0
rec1 = account(m_rec)            # recipient account #1 (SAME identity)
M_rec = mul(G1, m_rec)           # what the issuer learns out of band

tree = IdentityTree()             # unsalted points: what a receipt checks
for a in (issuer, rec0):          # one leaf per *identity*
    tree.insert(a["M"])

# The mailbox is not the Identity. k comes from the wallet seed, and the
# private registry commits the pair (m_rec, k) under the holder's salt.
# The issuer's SECOND association is an ordinary salted identity leaf: its
# salt is what an A2 payload ships, and it says nothing about the issuer's
# own mailbox. The registry's subtree is enrolled under an aggregator, so
# its paths are the 32-level paths the gates prove: 12, then 20.
KYC = "kyc:ca-ab-2026"
seed_rec = rand_scalar(rng)
k_rec, pk_recv = receiving_key(seed_rec)
salt_rec = derive_salt(seed_rec, KYC)
seed_iss = rand_scalar(rng)
salt_iss = derive_salt(seed_iss, KYC, 1)
priv = rooted_registry(KYC)
priv.insert_leaf(identity_leaf_salted(issuer["M"], salt_iss))
priv.insert_receiving(m_rec, k_rec, salt_rec)

print(f"issuer Identity    M_I = {pt(issuer['M'])}")
print(f"recipient Identity M_rec = {pt(M_rec)} (accounts rec0, rec1)")
print(f"receiving key      pk_recv = {pt(pk_recv)} (not M_rec)")
print(f"identity tree root   = {hex(tree.root())[20]}...")
print(f"posted identity root = {hex(priv.root())[20]}... ({priv.depth}-level paths)")

wallet_crypto_backend = kernel (buck_core.buck_identity)

```

```

issuer Identity    M_I    = (34794633..., 55370756...)
recipient Identity M_rec = (78777287..., 1501867...) (accounts rec0, rec1)
receiving key      pk_recv = (27820058..., 70606247...) (not M_rec)
identity tree root = 0xff9f26af3a3e77cf95...
posted identity root = 0x638a64903f90cc47d9... (32-level paths)

```

Mint: encrypt the issuer's own Identity under the recipient's mailbox. The issuer addresses the note to the Identity and keys both ciphertexts to pk_{recv} , learned on the same out-of-band channel that carries M_{rec} . It needs nothing from the recipient at payment time. The anti-framing binding (`verifyIssuerReenc`, with the recipient slot set to pk_{recv}) proves $eIss$ re-encrypts the issuer's own registered M_I . The mint object carries $salt_{\text{iss}}$, the salt of the issuer's naming association, and r' , the randomness of $eIss$. The fold checks $eIss.R = r'G$ and opens `identity_leaf_salted( $M_I, salt_{\text{iss}}$ )`; neither value is the issuer's mailbox salt. The binding's T is committed in `idHash` beside both ciphertexts, and its blind γ is the third thing the recipient needs. The delivery (`wallet/delivery.py`) carries r' , γ and $salt_{\text{iss}}$ wrapped to pk_{recv} , with T in the clear.

```

note = mint_unilateral_a2(issuer["sk"], issuer["E"], pk_recv,
                          v=1000, rho=rand_scalar(rng),
                          issuer=0xA11CE, chainid=CHAINID,
                          salt_iss=salt_iss, rng=rng)
assert eq(note.M_I, issuer["M"])
assert note.salt_iss == salt_iss
print(f"minted A2 note: v = {note.opening.v}")
print(f"  cm      = {hex(note.cm)[:20]}...      (the only on-chain artifact)")
print(f"  idHash = {hex(note.idHash)[:20]}...    (Poseidon10(eNote, eIss, T))")
print(f"  eIss    = ElGamal(M_I under pk_recv) R = {pt(note.eIss.R)}")

```

```

minted A2 note: v = 1000
  cm      = 0xe61242769fe57dca10...      (the only on-chain artifact)
  idHash = 0xf22f5585ad9a988d15...    (Poseidon10(eNote, eIss, T))
  eIss    = ElGamal(M_I under pk_recv) R = (49017840..., 29504036...)

```

Delivery: the recipient names the issuer with the receiving secret. The decryption secret is k , not m_{rec} . Every account the recipient owns shares m_{rec} ; none of them opens the note.

$$C_e - k R_e = (M_I + r' pk_{\text{recv}}) - k(r'G) = M_I.$$

```

M_I = elgamal_decrypt(note.eIss, k_rec)
assert eq(M_I, issuer["M"])
opened_by_m = eq(elgamal_decrypt(note.eIss, m_rec), issuer["M"])
print("recipient decrypts eIss with k -> the issuer's registered Identity:")
print(f"  C_e - k*R_e = {pt(M_I)} == M_I")
print(f"  the identity scalar opens it: {opened_by_m}")

```

```

recipient decrypts eIss with k -> the issuer's registered Identity:
  C_e - k*R_e = (34794633..., 55370756...) == M_I
  the identity scalar opens it: False

```

Deposit from ANY account bound to m_{rec} . Authority is the Identity, so both `rec0` and `rec1` spend the same note under the one mailbox key. The gate is one witness. It refuses unless k decrypts the ciphertext, the account credential decrypts to M_{rec} , and a registered leaf commits the pair (m_{rec}, k) under the holder's salt, on a path to the posted root. There is no shared public point for a second proof to meet at.

```

for name, acct in (("rec0", rec0), ("rec1", rec1)):
    w_acct = deposit_fold_witness(
        m_rec=m_rec, k=k_rec, sk_dep=acct["sk"], salt=salt_rec,
        E_dep=acct["E"], note_ct=note.eIss, tree=priv)
    ok = deposit_fold_check(
        w_acct, pk_dep=acct["pk"], E_dep=acct["E"],
        note_ct=note.eIss, root=priv.root())
    print(f"folded gate from {name}: verifies = {ok}")

```

```

folded gate from rec0: verifies = True
folded gate from rec1: verifies = True

```

On chain the same statement is one Groth16 proof. `DepositFoldVerifierAdapter.verifyFoldA2` derives every public input: the nullifier, the posted identity root, the ciphertext, and the depositor's registered key and credential, the last two read from the registry rather than accepted from the caller. The proof bytes are only the Groth16 triple, and they are the whole of the addressed calldata beside the re-encryption:

```

// Notes._spendCoupled (abridged).
require(identityRegistry.acceptsRoot(identityRoot, NOTES_MEMBERSHIP_CONSUMER),
    "Notes: identity root not accepted");
bool ok = depositFoldVerifier.verifyFoldA2(
    foldProof, nullifier, identityRoot, eEnc, msg.sender);
require(ok, "Notes: bad folded deposit gate");

```

The fold witness: the note tie and the issuer's naming leaf, in one proof. The spend supplies a fresh re-encryption of $eIss$ under the same pk_{recv} . `deposit_fold_a2_witness` is the JSON the circuit consumes. It asserts the four relations above, the fifth – the decrypted issuer Identity opens `identity_leaf_salted(M_I, salt_iss)`, and that leaf's path folds to the same root – and the sixth, T opening against k . A wrong salt is not a member.

```

s = rand_scalar(rng)
t = (note.r_prime + s) % ORDER
eEnc = elgamal_encrypt(note.M_I, pk_recv, t)
w = deposit_fold_witness(
    m_rec=m_rec, k=k_rec, sk_dep=rec0["sk"], salt=salt_rec,
    E_dep=rec0["E"], note_ct=note.eIss, tree=priv)
iss_leaf = identity_leaf_salted(issuer["M"], salt_iss)
iss_path = priv.path(priv.leaves.index(iss_leaf))
fold = deposit_fold_a2_witness(
    witness=w, rho=note.opening.rho, id_hash=note.idHash,
    e_note=note.eNote, e_iss=note.eIss, r_prime=note.r_prime,
    t=t, r_E=rec0["r"], e_dep=rec0["E"], pk_dep=rec0["pk"],
    e_enc=eEnc, salt_iss=salt_iss, iss_path=iss_path,
    T=note.binding.T, gamma=note.gamma, identity_root=priv.root())
print(f"A2 fold witness built: {len(fold)} signals, relations asserted")
print(f" public nullifier = {hex(int(fold['nullifier']))[:20]}...")
print(f" eEnc.R limb0 = {fold['eEncRx'][0]} (fresh re-encryption)")
assert fold["saltIss"] == str(salt_iss)
assert int(fold["eEncRx"][0]) != 0

try:
    deposit_fold_a2_witness(
        witness=w, rho=note.opening.rho, id_hash=note.idHash,
        e_note=note.eNote, e_iss=note.eIss, r_prime=note.r_prime,
        t=t, r_E=rec0["r"], e_dep=rec0["E"], pk_dep=rec0["pk"],
        e_enc=eEnc, salt_iss=(salt_iss + 1) % ORDER, iss_path=iss_path,
        T=note.binding.T, gamma=note.gamma, identity_root=priv.root())
    print("wrong issuer salt: BUILT (must not happen!)")
except AssertionError as exc:
    print(f"wrong issuer salt: REFUSED ({exc})")

```

```

A2 fold witness built: 34 signals, relations asserted
public nullifier = 0x15373dec2a263756e1...
eEnc.R limb0    = 5454261353227722156 (fresh re-encryption)
wrong issuer salt: REFUSED (the shipped issuer salt does not open the issuer's leaf)

```

Proving this witness is the heavyweight step – 3,833,335 constraints along 32-level paths – and it runs in the build pipeline (`scripts/snark/setup_deposit_fold_a2.sh`) rather than in this session. The setups are dev entropy. `test/DepositFoldA2Verifier.t.sol` verifies a real proof on chain and requires every public input, walked alone, to break it.

The unilateral receipt: the recipient names BOTH parties, alone. From k and its own Identity – two values, not one derived from the other – the recipient recovers M_I and produces a third-party-checkable receipt. The verifiable decryption is under pk_{recv} . The membership checks below are membership of the two Identity *points* in the unsalted tree the receipt was given. They are not the spend's receiving-leaf check; that check is the fold.

```

receipt = make_receipt(k_rec, M_rec, note, issuer=0xA11CE, chainid=CHAINID,
                      tree=tree, rng=rng)
res = verify_receipt(receipt, issuer["pk"], issuer["E"], tree.root(), tree)
assert res.valid
assert eq(res.issuer_M, issuer["M"])
assert eq(res.recipient_M, M_rec)
assert res.value == 1000
assert eq(receipt.pk_recv, pk_recv)
print(f"unilateral receipt: {res.reason}")
print(f"  names issuer      {pt(res.issuer_M)}")
print(f"  names recipient  {pt(res.recipient_M)}")
print(f"  decrypts under    {pt(receipt.pk_recv)}")
print(f"  value              {res.value}")

```

```

unilateral receipt: VALID
names issuer      (34794633..., 55370756...)
names recipient  (78777287..., 1501867...)
decrypts under   (27820058..., 70606247...)
value            1000

```

A stolen payload does not spend, and a throwaway key does not name. A thief holding k and its own registered Identity satisfies each half of a split gate. The fold refuses on the leaf: no registered leaf pairs the thief's Identity with the key it stole. Separately, a ciphertext keyed to a key the recipient does not hold decrypts to garbage, and that point is not a registered Identity, so the receipt does not verify.

```

m_thief = rand_scalar(rng)
thief = account(m_thief)
seed_t = rand_scalar(rng)
k_t, _pk_t = receiving_key(seed_t)
salt_t = derive_salt(seed_t, KYC)
priv.insert_receiving(m_thief, k_t, salt_t)
try:
    deposit_fold_witness(
        m_rec=m_thief, k=k_rec, sk_dep=thief["sk"], salt=salt_t,
        E_dep=thief["E"], note_ct=note.eIss, tree=priv)
    print("thief witness: BUILT (must not happen!)")
except DepositFoldRefused as exc:
    print(f"thief witness: REFUSED on relation {exc.relation}")

pk_bogus = mul(G1, rand_scalar(rng))
eIss_bogus = elgamal_encrypt(issuer["M"], pk_bogus, rand_scalar(rng))
M_I_bogus = elgamal_decrypt(eIss_bogus, k_rec)
print(f"ciphertext under a throwaway key decrypts to {pt(M_I_bogus)}")
print(f"that point is a registered identity: {tree.contains(M_I_bogus)}")

```

```

thief witness: REFUSED on relation 3
ciphertext under a throwaway key decrypts to (49911686..., 9395411...)
that point is a registered identity: False

```

A split key names no puppet. The mint binding proves $eIss$ carries the minter's own Identity to the key hidden in Q , and an ElGamal ciphertext does not commit to its key. So a minter holding a registered sock puppet M_B picks $pk_{\text{mint}} = pk_{\text{recv}} + r'^{-1}(M_B - M_I)$: the same $eIss$ is its own M_I under pk_{mint} and M_B under Bob's key. The binding verifies, relation (5) holds on M_B 's leaf, and without more an honest Bob would name the puppet. The more is T : the note commits it, and it opens under the key the minter chose, not Bob's. Bob's wallet refuses the delivery, the fold refuses relation (6), and the receipt refuses to name M_B . `mint_unilateral_a2` cannot produce this note, so it is minted by hand:

```

from alberta_buck.wallet.issuer_reenc import issuer_reenc_prove, issuer_reenc_verify
from alberta_buck.wallet.unilateral_a2 import MintedA2, a2_id_hash
from alberta_buck.wallet.notes import FLAVOR_A2, NoteOpening, note_commitment
from alberta_buck.wallet.delivery import DeliveryRefused, deliver_a2, open_a2

_seed_sp = random.Random(0x5B1175)          # local; the shared transcript is untouched
rng_sp   = lambda: _seed_sp.getrandbits(256)

M_B      = mul(G1, rand_scalar(rng_sp))      # the minter's registered sock puppet
salt_B   = derive_salt(rand_scalar(rng_sp), KYC, 1)
priv.insert_identity_salted(M_B, salt_B)
tree.insert(M_B)

r_s      = rand_scalar(rng_sp)
pk_mint  = add(pk_recv, mul(add(M_B, neg(issuer["M"])), pow(r_s, -1, ORDER)))
eIss_s   = elgamal_encrypt(issuer["M"], pk_mint, r_s)
gamma_s  = rand_scalar(rng_sp)
bind_s   = issuer_reenc_prove(issuer["sk"], r_s, pk_mint, issuer["E"], eIss_s,
                              0xA11CE, CHAINID, gamma=gamma_s, rng=rng_sp)
rn_s     = rand_scalar(rng_sp)
eNote_s  = elgamal_encrypt(mul(G1, 1000), pk_recv, rn_s)
idh_s    = a2_id_hash(eNote_s, eIss_s, bind_s.T)
open_s   = NoteOpening(FLAVOR_A2, 1000, rand_scalar(rng_sp), idh_s, 0)
split    = MintedA2(eNote=eNote_s, eIss=eIss_s, M_I=issuer["M"], idHash=idh_s,
                   cm=note_commitment(open_s), opening=open_s, binding=bind_s,
                   r_prime=r_s, r_note=rn_s, gamma=gamma_s, salt_iss=salt_B)

print(f"binding verifies against Alice's record: "
      f"{issuer_reenc_verify(issuer['pk'], issuer['E'], eIss_s, bind_s, 0xA11CE, CHAINID)}")
print(f"Bob's k opens eIss to the puppet:      {eq(elgamal_decrypt(eIss_s, k_rec), M_B)}")
try:
    open_a2(deliver_a2(split, pk_recv), k_rec)
    print("delivery: OPENED (must not happen!)")
except DeliveryRefused as exc:
    print(f"delivery: REFUSED ({exc})")

t_s      = (r_s + rand_scalar(rng_sp)) % ORDER
eEnc_s   = elgamal_encrypt(M_B, pk_recv, t_s)
w_s      = deposit_fold_witness(m_rec=m_rec, k=k_rec, sk_dep=rec0["sk"], salt=salt_rec,
                               E_dep=rec0["E"], note_ct=eEnc_s, tree=priv)
try:
    deposit_fold_a2_witness(
        witness=w_s, rho=open_s.rho, id_hash=idh_s, e_note=eNote_s, e_iss=eIss_s,
        r_prime=r_s, t=t_s, r_E=rec0["r"], e_dep=rec0["E"], pk_dep=rec0["pk"],
        e_enc=eEnc_s, salt_iss=salt_B,
        iss_path=priv.path(priv.leaves.index(identity_leaf_salted(M_B, salt_B))),
        T=bind_s.T, gamma=gamma_s, identity_root=priv.root())
    print("fold: BUILT (must not happen!)")
except AssertionError as exc:
    print(f"fold: REFUSED ({exc})")

```

```

res_s = verify_receipt(make_receipt(k_rec, M_rec, split, issuer=0xA11CE, chainid=CHAINID,
                                   tree=tree, rng=rng_sp),
                      issuer["pk"], issuer["E"], tree.root(), tree)
print(f"receipt: {'VALID' if res_s.valid else 'INVALID'} ({res_s.reason})")

```

```

binding verifies against Alice's record: True
Bob's k opens eIss to the puppet: True
delivery: REFUSED (T does not open to this mailbox under the unwrapped gamma)
fold: REFUSED (T != rm*G + gamma*H (keyed to another mailbox))
receipt: INVALID (the binding's Identity is not the one decrypted)

```

That is the guarantee: the recipient holds the secret that names the issuer, the identity scalar does not, a stolen mailbox key does not spend under a different Identity, and a key chosen to frame a puppet does not name one.

12.4 The Receiving Key: Your Identity Is Not Your Mailbox Key

An addressed note names an Identity and is keyed to that Identity's *receiving key*. Those are two different objects doing two different jobs, and it is worth being clear why, because collapsing them into one is the most natural design in the world and it does not work.

The read/write split makes m a *read capability*: the design hands it to every counterparty, because that is how a receipt names a person. So it cannot also be the thing that opens the recipient's mail. One disclosure would read every note that person ever receives, in both directions in time. A read capability you give away cannot be a decryption key you keep.

The arithmetic is even less forgiving than the principle. Suppose a leaf encrypted the recipient's Identity *under itself*:

$$eRec = (r'G, M_{\text{rec}} + r'M_{\text{rec}}) = (r'G, m_{\text{rec}}(G + R)).$$

Message and key would share one secret, so the Diffie-Hellman hardness protecting every other ciphertext here would simply be absent. A candidate Identity tests with *one* scalar multiplication – no proof, no key, no issuer secret – and an issuer holding its client list could ask "is any note in existence addressed to anyone I certified?" and get answers.

Keying the same plaintext to pk_{recv} instead turns that question into a decisional Diffie-Hellman decision, which is the lock the rest of this design already rests on. Run both and watch the difference:

```

import random
from alberta_buck.wallet.elgamal import elgamal_encrypt, elgamal_decrypt
from alberta_buck.wallet.salt import derive_salt
from alberta_buck.wallet.deposit_fold import (
    DepositFoldRefused, deposit_fold_witness,
)
from alberta_buck.registry.tree import IdentityMerkleTree

_seed_rk = random.Random(0xBADDECAF) # local; the shared transcript is untouched
rng_rk = lambda: _seed_rk.getrandbits(256)

m_rec_rk = rand_scalar(rng_rk) # the recipient's Identity
M_rec_rk = mul(G1, m_rec_rk)
k_rk = rand_scalar(rng_rk) # the recipient's RECEIVING key
pk_demo = mul(G1, k_rk) # local; must not clobber the session's pk_recv

# A harvester's candidate list: every Identity it ever certified, plus the one
# it is hunting. It holds no keys and needs none.
cands = [rand_scalar(rng_rk) for _ in range(8)] + [m_rec_rk]

# Today's A1 leaf ciphertext: the Identity encrypted under ITSELF.

```

```

eRec_today = elgamal_encrypt(M_rec_rk, M_rec_rk, rand_scalar(rng_rk))
hit = [c for c in cand_s if eq(eRec_today.C, mul(add(G1, eRec_today.R), c))]
print(f"today: C == m*(G+R) over {len(cand_s)} candidates -> {len(hit)} hit"
      f" ({'IDENTIFIED' if hit == [m_rec_rk] else 'no'})")

# The same plaintext under a key that is not the Identity.
eRec_new = elgamal_encrypt(M_rec_rk, pk_demo, rand_scalar(rng_rk))
hit2 = [c for c in cand_s if eq(eRec_new.C, mul(add(G1, eRec_new.R), c))]
print(f"fixed: the same scan over the same candidates -> {len(hit2)} hit")
print(f" recipient still reads it with k: {eq(elgamal_decrypt(eRec_new, k_rk), M_rec_rk)}")
print(f" the Identity scalar no longer does: {eq(elgamal_decrypt(eRec_new, m_rec_rk), M_rec_rk)}")

today: C == m*(G+R) over 9 candidates -> 1 hit (IDENTIFIED)
fixed: the same scan over the same candidates -> 0 hit
       recipient still reads it with k: True
       the Identity scalar no longer does: False

```

Nine candidates, one hit under the collapsed design, zero under the real one. The recipient still reads its mail; it reads it with k , a secret it hands to nobody.

Two secrets need a gate that joins them. Separating reading from Identity buys the privacy above and creates an obligation: the deposit gate must say something a single-secret design would get for free. "I can read this note" and "I am this registered Identity" are facts about two different secrets, and a gate that proved them side by side would say nothing at all about their owner:

```

sk_rk = rand_scalar(rng_rk); pk_rk = mul(G1, sk_rk)
E_dep_rk = elgamal_encrypt(M_rec_rk, pk_rk, rand_scalar(rng_rk)) # registered account
m_iss_rk = rand_scalar(rng_rk); M_I_rk = mul(G1, m_iss_rk)
eIss_rk = elgamal_encrypt(M_I_rk, pk_demo, rand_scalar(rng_rk)) # A2 note, to the demo key

def split_gate(m_id, k_read, sk_acct, E_acct, ct):
    """The deposit gate with the two jobs merely separated: prove the account
    holds m_id, and prove k_read decrypts the note. Nothing joins them."""
    holds = eq(E_acct.C, add(mul(G1, m_id), mul(E_acct.R, sk_acct)))
    M = add(ct.C, neg(mul(ct.R, k_read))) # whatever k decrypts to
    return holds, M

holds, M = split_gate(m_rec_rk, k_rk, sk_rk, E_dep_rk, eIss_rk)
print(f"honest deposit: account holds the Identity = {holds}, "
      f"decrypts to M_I = {eq(M, M_I_rk)}")

m_thief = rand_scalar(rng_rk); sk_thief = rand_scalar(rng_rk)
E_thief = elgamal_encrypt(mul(G1, m_thief), mul(G1, sk_thief), rand_scalar(rng_rk))
holds_t, M_t = split_gate(m_thief, k_rk, sk_thief, E_thief, eIss_rk)
print(f"thief with a stolen payload and its OWN Identity:")
print(f" its account holds its own Identity = {holds_t}")
print(f" the stolen k decrypts to M_I = {eq(M_t, M_I_rk)}")
print(f" -> both halves pass. The note is spendable by the wrong person.")

honest_deposit: account holds the Identity = True, decrypts to M_I = True
thief with a stolen payload and its OWN Identity:
  its account holds its own Identity = True
  the stolen k decrypts to M_I = True
  -> both halves pass. The note is spendable by the wrong person.

```

A thief holding a stolen payload proves the reading half with the stolen k and the Identity half with its *own* registered Identity. Both halves are true. Neither says they belong together.

So the gate states the tie instead of assuming it – the discipline every relation in the folded gates follows, and for the same reason: *never infer equality from two proofs that happen to share a public point*. One witness, one circuit, four relations, with the accumulator leaf committing the scalars $(m, k, salt)$:

```

secret_rk = rand_scalar(rng_rk)
salt_rk = derive_salt(secret_rk, "kyc:ca-ab-2026")
tree_rk = IdentityMerkleTree(depth=10, private=True)
tree_rk.insert_receiving(m_rec_rk, k_rk, salt_rk)          # scalars, not points
tree_rk.insert_receiving(m_thief, rand_scalar(rng_rk),
                        derive_salt(rand_scalar(rng_rk), "kyc:ca-ab-2026"))

# deposit_fold_witness carries the four relations over ONE witness and
# refuses on the one that fails. It is the reference the circuit is
# checked against, so this block runs the shipped gate, not a sketch.
w = deposit_fold_witness(m_rec=m_rec_rk, k=k_rk, sk_dep=sk_rk, salt=salt_rk,
                        E_dep=E_dep_rk, note_ct=eIss_rk, tree=tree_rk)
print(f"honest witness: exists, k decrypts to M_I = {eq(w.M, M_I_rk)}")
try:
    deposit_fold_witness(m_rec=m_thief, k=k_rk, sk_dep=sk_thief, salt=salt_rk,
                        E_dep=E_thief, note_ct=eIss_rk, tree=tree_rk)
    print("thief witness: BUILT (must not happen!)")
except DepositFoldRefused as exc:
    print(f"thief witness: REFUSED on relation {exc.relation}")
k2 = rand_scalar(rng_rk); salt2 = derive_salt(secret_rk, "kyc:ca-ab-2026", 1)
tree_rk.insert_receiving(m_rec_rk, k2, salt2)
w2 = deposit_fold_witness(m_rec=m_rec_rk, k=k2, sk_dep=sk_rk, salt=salt2,
                        E_dep=E_dep_rk, tree=tree_rk,
                        note_ct=elgamal_encrypt(M_I_rk, mul(G1, k2),
                                                rand_scalar(rng_rk)))
print(f"rotated key: spends, under an unlinkable leaf = {w2.leaf != w.leaf}")

honest witness: exists, k decrypts to M_I = True
thief witness: REFUSED on relation 3
rotated key: spends, under an unlinkable leaf = True

```

The thief fails on relation (3): no registered leaf pairs *its* Identity with the key it stole. Rotation works – a fresh receiving key, a fresh salt, a leaf that links to nothing – because rotation is the accumulator’s re-association.

Where each secret lives, and why that is the right place. Authority is the Identity, so losing an account key strands nothing: register another account on the same Identity and deposit. Reading is a key, so it lives in the wallet seed, which is where every wallet has always kept reading keys, with guardian-wrapped copies for the cases that warrant more. What the Identity scalar deliberately cannot do is reconstruct the reading key, because a key derivable from a record you have disclosed is a key held by everyone who has the record. The cost of naming yourself to a counterparty is that they can name you back. It stops there.

What the separation does not cost. Addressing stays unilateral: the payer needs pk_{recv} , learned on the same out-of-band channel that carries M_{rec} , and needs nothing from the recipient at payment time. Receipts still name both parties. The note tie keeps its shape, since it witnesses a product of the randomness and the recipient’s scalar and computes it fixed-base, and $r \cdot k$ is as fixed-base as anything else. And B1 needs none of this: a bearer note is keyed to the *issuer’s* registered account key, so the question never arises.

Where to read further. The shipped surface is `wallet/recvkey.py` (the key, its seed derivation and its binding evidence), `registry/tree.py` `receiving_leaf` (the leaf that commits the pair) and `wallet/deposit_fold.py` (the gate above), with twenty-five tests in `alberta_buck/test/test_receiving_key`. The same arguments also run standalone in `scripts/review/receiving_key_proto.py`, `deposit_gate_split.py` and `deposit_gate_folded.py`.

12.5 A1 in Code: The Public-Issuer Addressed Cycle

A1 is A2 with the issuer in the clear. *idHash* commits $(eNote, m_{\text{issuer}}, \sigma)$ instead of a second ciphertext. *eRec* encrypts the plaintext M_{rec} under pk_{recv} , not under M_{rec} : the Identity is what the

note names, and the mailbox key is what opens it. The spend gate is `deposit_fold_a1`, the same four relations as A2 without the issuer-membership relation, because the plaintext of `eRec` is the recipient's own Identity, which the receiving leaf already commits. The face v is a public input of that proof. (The batch Schnorr is checked at the mint transaction – `verifyIssuerSchnorr`; here the signature is synthetic, since the session has no chain.)

```
from alberta_buck.wallet.unilateral_a1 import (
    mint_unilateral_a1, make_receipt_a1, verify_receipt_a1,
)
from alberta_buck.wallet.deposit_fold import deposit_fold_a1_witness

k_sig = rand_scalar(rng)
sigma_R, sigma_s = mul(G1, k_sig), (k_sig + rand_scalar(rng) * rand_scalar(rng)) % ORDER

note_a1 = mint_unilateral_a1(M_rec, pk_recv, v=100, rho=rand_scalar(rng),
                             m_issuer=m_iss, sigma_R=sigma_R, sigma_s=sigma_s,
                             rng=rng)
print(f"minted A1 note: v = {note_a1.opening.v}")
print(f" idHash = Poseidon8(eNote, m_issuer, sigma) = {hex(note_a1.idHash)[:20]}...")
print(f" eRec = ElGamal(M_rec under pk_recv) R = {pt(note_a1.eRec.R)}")
print(f" identity scalar opens eRec: "
      f"{eq(elgamal_decrypt(note_a1.eRec, m_rec), M_rec)}")
print(f" receiving secret opens eRec: "
      f"{eq(elgamal_decrypt(note_a1.eRec, k_rec), M_rec)}")

rcpt_a1 = make_receipt_a1(k_rec, M_rec, note_a1, issuer["M"], issuer=0xA11CE,
                          chainid=CHAINID, tree=tree, rng=rng)
res_a1 = verify_receipt_a1(rcpt_a1, tree.root(), tree)
assert res_a1.valid and res_a1.value == 100
assert eq(rcpt_a1.pk_recv, pk_recv)
print(f"A1 receipt: {res_a1.reason} -- issuer public, decryption under pk_recv")

minted A1 note: v = 100
idHash = Poseidon8(eNote, m_issuer, sigma) = 0x15db44b7e86fcdef0c...
eRec = ElGamal(M_rec under pk_recv) R = (92561594..., 14543584...)
identity scalar opens eRec: False
receiving secret opens eRec: True
A1 receipt: VALID -- issuer public, decryption under pk_recv
```

The A1 fold witness: the value ciphertext is pinned by the public face. `deposit_fold_a1_witness` is what `deposit_fold_a1.circom` consumes. It checks $eNote = (r_n G, (v + r_n k)G)$ and $eEnc = (tG, (m_{rec} + tk)G)$, with v public, and the same leaf $Poseidon(\tau_R, m_{rec}, k, salt)$. A different receiving secret does not open $eNote$. A thief that holds k and substitutes its own Identity is refused by the leaf, the same relation the A2 block already showed.

```
w_a1 = deposit_fold_witness(
    m_rec=m_rec, k=k_rec, sk_dep=rec0["sk"], salt=salt_rec,
    E_dep=rec0["E"], note_ct=note_a1.eRec, tree=priv)
t_a1 = rand_scalar(rng)
eEnc_a1 = elgamal_encrypt(M_rec, pk_recv, t_a1)
fold_a1 = deposit_fold_a1_witness(
    witness=w_a1, rho=note_a1.opening.rho, id_hash=note_a1.idHash,
    e_note=note_a1.eNote, v=note_a1.opening.v, m_issuer=m_iss,
    sigma_R=sigma_R, sigma_s=sigma_s, r_note=note_a1.r_note,
    t=t_a1, r_E=rec0["r"], e_dep=rec0["E"], pk_dep=rec0["pk"],
    e_enc=eEnc_a1, identity_root=priv.root())
print(f"A1 fold witness built: {len(fold_a1)} signals, relations asserted")
print(f" public face v = {fold_a1['v']}")
print(f" public nullifier = {hex(int(fold_a1['nullifier']))[:20]}...")

k_other, _pk_other = receiving_key(rand_scalar(rng))
try:
```

```

bad = deposit_fold_witness(
    m_rec=m_rec, k=k_other, sk_dep=rec0["sk"], salt=salt_rec,
    E_dep=rec0["E"], note_ct=note_a1.eRec, tree=priv)
print(f"wrong mailbox key: BUILT (must not happen!) leaf={bad.leaf}")
except DepositFoldRefused as exc:
    print(f"wrong mailbox key: REFUSED on relation {exc.relation}")

```

```

A1 fold witness built: 31 signals, relations asserted
public face v      = 100
public nullifier = 0x1c454e5d6eec110e7c...
wrong mailbox key: REFUSED on relation 3

```

The circuit is 3,309,327 constraints on a 32-level path. `DepositFoldA1Verifier` verifies a real proof on chain (`test/DepositFoldA1Verifier.t.sol`), and `test/NotesE2E.t.sol` spends an A1 note through the fold inside the full lifecycle. The dev setup is not a ceremony.

12.6 B1 Bearer Spend with Depositor Naming (The A2 Dual)

Carol holds a B1 note from a public issuer. She has never been identified to the issuer; at spend she reveals her Identity to the issuer *alone*, through an encryption only the issuer can decrypt.

==== Step B0: Carol Holds a B1 Note ====

Carol's artifact carries the issuer's public Identity M_{issuer} in the clear with a Schnorr signature, but *no* per-recipient ciphertext (bearer semantics).

==== Step B1: Carol Encrypts Her Identity for the Issuer ====

At deposit Carol's wallet encrypts her Identity under the public issuer's key:

$$E_{\text{dep} \rightarrow \text{iss}} = (R_f, C_f) = (rG, M_{\text{Carol}} + r pk_{\text{iss}}).$$

Only sk_{iss} recovers M_{Carol} ; to Mallory the ciphertext is IND-CPA noise.

==== Step B2: Generate the Depositor-Binding Proof (A2 dual) ====

Carol's wallet proves (`b1_bind_prove`, `verifyDepositorBinding`) knowledge of $(m_{\text{Carol}}, sk_{\text{dep}}, r)$ such that the ciphertext encrypts the *same* Identity bound to Carol's payout account:

$$\begin{array}{ll}
 \text{(E4)} & pk_{\text{dep}} = sk_{\text{dep}} G; \\
 \text{(E2)} & C_d = m_{\text{Carol}} G + sk_{\text{dep}} R_d; \\
 \text{(F1)} & R_f = r G; \\
 \text{(F2)} & C_f = m_{\text{Carol}} G + r pk_{\text{iss}}; \\
 \text{(P)} & P_{\text{dep}} = m_{\text{Carol}} G + b H.
 \end{array}$$

The shared witness m_{Carol} couples (E2), (F2), and (P): the Identity bound to Carol's payout account is exactly the one encrypted for the issuer *and* the one committed (blinded) in P_{dep} – so a colluding depositor cannot substitute a different Identity, and the membership below is bound to the right point. (11 Python + 6 Solidity tests.)

==== Step B3: Submit `Notes.spendCoupledB1` ====

Carol calls the membership-bound B1 deposit:

```

notes.spendCoupledB1(
    proof, noteRoot, identityRoot, nullifier, face, recipient,
    issuanceCommitment, // the mint commitment the spend opened
    issuer, // public issuer address
    eDepForIss, // (R_f, C_f) encrypting M_Carol
    b1Proof, // verifyDepositorBinding (incl. P_dep)
    membershipProof, // membership of M_Carol, bound to P_dep
);

```

The contract verifies the binding, then the membership of `b1Proof.P_dep` (the *same* point the binding committed), and emits `SpentCoupledB1(nullifier, face, recipient, issuer, eDepForIss)` with the ciphertext only the issuer can decrypt. Carol is now provably a registered Identity the issuer can name – the membership-bound dual of the A2 recipient receipt (`test/NotesCoupledB1.t.sol`, 6 tests).

==== Step B4: Issuer Decrypts and Produces a Receipt ====

The issuer scans `SpentCoupledB1` events, decrypts M_{Carol} with sk_{iss} , and *alone* produces a plaintext receipt naming both parties:

$$\mathcal{R}_{\text{iss}} = (M_{\text{issuer}}, M_{\text{Carol}}, v, E_{\text{dep} \rightarrow \text{iss}}, \text{vd}, \text{chainid}).$$

(make_issuer_receipt in alberta_buck.wallet.b1_binding.)

12.7 B1 in Code: The Issuer Names the Depositor

The same session, continued. Carol is a registered Identity the (public) issuer has never met; she holds a bearer note's opening. At spend she builds `eDepForIss` – her own M_{dep} encrypted under the issuer's key – and the five-relation depositor-binding sigma couples it to her payout account and to the blinded membership commitment P_{dep} :

```
from alberta_buck.wallet.b1_binding import (
    b1_bind_prove, b1_bind_verify, make_issuer_receipt, verify_issuer_receipt,
)

m_carol = rand_scalar(rng)
carol = account(m_carol)
tree.insert(carol["M"])

pk_iss = issuer["pk"]
binding, eDepForIss = b1_bind_prove(m_carol, carol["sk"], carol["E"], pk_iss,
                                   account=0xCA801, chainid=CHAINID, rng=rng)
ok = b1_bind_verify(carol["pk"], carol["E"], pk_iss, eDepForIss, binding,
                   account=0xCA801, chainid=CHAINID)
print(f"B1 depositor binding from Carol: verifies = {ok}")
print(f"  eDepForIss = ElGamal(M_carol under pk_iss)  R = {pt(eDepForIss.R)}")
print(f"  P_dep      = {pt(binding.P_dep)} (blinded; membership binds it)")

B1 depositor binding from Carol: verifies = True
eDepForIss = ElGamal(M_carol under pk_iss)  R = (62881448..., 51371453...)
P_dep      = (87656785..., 92071539...) (blinded; membership binds it)
```

The issuer alone – reading `eDepForIss` from the `SpentCoupledB1` event – decrypts and *proves* the depositor's Identity, the dual of the A2 unilateral receipt. And the collusion-substitution is rejected: a ciphertext under any other key breaks the coupled sigma, so the depositor cannot mis-name itself.

```
rcpt_iss = make_issuer_receipt(issuer["sk"], issuer["M"], eDepForIss,
                              value=25, issuer=0xA11CE, chainid=CHAINID,
                              tree=tree, rng=rng)
res_b1 = verify_issuer_receipt(rcpt_iss, tree.root(), tree)
assert res_b1.valid
assert eq(res_b1.recipient_M, carol["M"])
print(f"issuer-unilateral receipt: {res_b1.reason}")
print(f"  names depositor {pt(res_b1.recipient_M)} == M_carol")

eDep_sub = elgamal_encrypt(carol["M"], mul(G1, rand_scalar(rng)), rand_scalar(rng))
ok_sub = b1_bind_verify(carol["pk"], carol["E"], pk_iss, eDep_sub, binding,
                      account=0xCA801, chainid=CHAINID)
print(f"substituted eDepForIss verifies = {ok_sub} (collusion-substitution rejected)")
```

```

issuer-unilateral receipt: VALID
  names depositor (87291686..., 3992517...) == M_carol
substituted eDepForIss verifies = False (collusion-substitution rejected)

```

12.8 Summary: What the Observer Sees (Mallory)

In all three flows the chain reveals:

- A fresh nullifier and a spent face amount.
- The payout address.
- For A1/A2: the re-encryption $eEnc$ under an unpublished pk_{recv} (semantically secure) and one fold proof; no committed point.
- For B1: the issuer address, $E_{\text{dep} \rightarrow \text{iss}}$ (IND-CPA) and the perfectly hiding P_{dep} .
- The posted identity root the spend names (one uint256 per posting, revealing nothing about which identities are active, since the leaves are salted).
- *No* plaintext Identities, *no* link between the spend and any specific commitment in the tree, and *no* link between mint and spend events.

13 The Prover Bill of Materials: Bandwidth, Storage, Time

Everything above *names* the proofs; this section prices them. The short answer to "is wallet-side proving expensive?" is **no** – **proving is seconds; distribution is the cost**. A Groth16 wallet never performs the trusted setup (the multi-GB Powers-of-Tau and the hours of zkey ceremony are a one-time, project-side event); it downloads two finished artifacts per circuit – the **proving key** (`.zkey`) and the **witness generator** (a wasm module, or a faster platform C++ binary plus its `.dat` constants) – and should pin their hashes against the published release. Verification costs a wallet *nothing* to distribute: the verifiers are contracts already on chain. And note the asymmetry that matters for UX: **receipts need no SNARK at all** – both parties' AB-RCPT/2 slips (Receipt) are sigma proofs and decryptions, milliseconds of wallet math – the SNARKs are paid only at mint and spend, once per note.

Who proves what:

- **Issuer, at mint:** one batch-mint proof – `mint_batch` (B1/A1) or `mint_batch_a2` (A2) at the chosen batch pin $N \in \{1, 2, 4, 8, 16, 32\}$ – plus artifact-free wallet sigmas (the batch Schnorr for B1/A1; the per-leaf `issuer_reenc` binding for A2; sub-millisecond each).
- **Depositor, at spend:** the flavor-agnostic `spend` proof, and then one identity proof. A1 and A2 prove the fold (`deposit_fold_a1` / `deposit_fold_a2`), which includes the note tie. B1 proves `identity_membership_b1` beside the depositor-binding sigma. Three keys, one per gate, and no others.
- **B1 is the light flavor:** no fold, and no addressed ciphertext at mint. Its membership key is the small one.

The blocks below are **live**: they read the artifact sizes from the local SNARK build (`build/snark/`, produced by `make snark` and friends) and then *actually run* every wallet prover once over the committed e2e fixture worlds, timing each. (The printed timing lines vary with hardware – Apple-silicon laptop here; every other line is deterministic.)

13.1 What the wallet downloads and stores

```

from pathlib import Path
SNARK = Path("build/snark")

def mb(*paths):
    total = 0
    for p in paths:
        pp = Path(p)
        if not pp.exists():
            return None
        total += pp.stat().st_size
    return total / 1e6

ARTIFACTS = [
    # circuit                consumer    zkey                witness generator (+ .dat)
    ("mint_batch N=1 (B1/A1 mint)", "issuer",
     ["mint_batch_n1/mint_batch_n1_final.zkey",
      ["mint_batch_n1/mint_batch_n1_js/mint_batch_n1.wasm"]]),
    ("mint_batch N=32 (B1/A1 mint)", "issuer",
     ["mint_batch_n32/mint_batch_n32_final.zkey",
      ["mint_batch_n32/mint_batch_n32_js/mint_batch_n32.wasm"]]),
    ("mint_batch_a2 N=1 (A2 mint)", "issuer",
     ["mint_batch_a2_n1/mint_batch_a2_n1_final.zkey",
      ["mint_batch_a2_n1/mint_batch_a2_n1_js/mint_batch_a2_n1.wasm"]]),
    ("mint_batch_a2 N=32 (A2 mint)", "issuer",
     ["mint_batch_a2_n32/mint_batch_a2_n32_final.zkey",
      ["mint_batch_a2_n32/mint_batch_a2_n32_js/mint_batch_a2_n32.wasm"]]),
    ("spend (all spends)", "depositor",
     ["spend/spend_final.zkey",
      ["spend/spend_js/spend.wasm"]]),
    ("b1 membership (B1 spend)", "depositor",
     ["b1_membership/b1m_0001.zkey",
      ["b1_membership/identity_membership_b1_js/identity_membership_b1.wasm"]]),
    ("deposit_fold_a1 (A1 spend)", "depositor",
     ["deposit_fold_a1/deposit_fold_a1_0001.zkey",
      ["deposit_fold_a1/deposit_fold_a1_cpp/deposit_fold_a1",
       ["deposit_fold_a1/deposit_fold_a1_cpp/deposit_fold_a1.dat"]]]),
    ("deposit_fold_a2 (A2 spend)", "depositor",
     ["deposit_fold_a2/deposit_fold_a2_0001.zkey",
      ["deposit_fold_a2/deposit_fold_a2_cpp/deposit_fold_a2",
       ["deposit_fold_a2/deposit_fold_a2_cpp/deposit_fold_a2.dat"]]]),
]
print(f"{'circuit':34s} {'consumer':10s} {'zkey':>10s} {'witness gen':>12s}")
print("-" * 70)
sizes = {}
for name, who, zkey, gens in ARTIFACTS:
    zs = mb(SNARK / zkey)
    gs = mb(*(SNARK / g for g in gens))
    if zs is None or gs is None:
        print(f"{name:34s} {who:10s} {'absent':>8s} {'absent':>12s}")
        continue
    sizes[name] = (zs, gs)
print(f"{'name':34s} {'who':10s} {zs:>8.1f}MB {gs:>10.1f}MB")

```

circuit	consumer	zkey	witness gen
mint_batch N=1 (B1/A1 mint)	issuer	10.0MB	2.6MB
mint_batch N=32 (B1/A1 mint)	issuer	319.6MB	5.3MB
mint_batch_a2 N=1 (A2 mint)	issuer	10.5MB	4.2MB
mint_batch_a2 N=32 (A2 mint)	issuer	335.5MB	7.1MB
spend (all spends)	depositor	5.5MB	2.9MB
b1 membership (B1 spend)	depositor	272.1MB	8.5MB
deposit_fold_a1 (A1 spend)	depositor	1863.4MB	36.0MB
deposit_fold_a2 (A2 spend)	depositor	2116.0MB	44.6MB

The two fold zkeys dominate. They are the proving keys for the 3,309,327-constraint A1 gate

and the 3,833,335-constraint A2 gate, and Groth16 proving keys scale with constraint count. B1's membership key is the 494,701-constraint circuit. Everything else a wallet fetches for a mint or a note opening is a few megabytes to a few hundred.

13.2 Running every wallet prover, timed

One real proof per circuit, over the same fixture worlds the live-EVM transcript spends (Receipt, *The Whole Flow, Executed*): snarkjs (wasm witness + prove) for the small circuits, the circom C++ witness generator

- rapidsnark for the big ties – exactly the pipeline

scripts/snark/gen_e2e_fixtures.sh runs.

```
import json, shlex, subprocess, tempfile, time

E2E = SNARK / "e2e"
FIX = Path("alberta_buck/test/vectors/e2e")
RAPID = next(Path("lib").glob("rapidsnark-*/bin/prover"))
tmp = Path(tempfile.mkdtemp(prefix="docflow-prove-"))

def timed(cmd):
    t0 = time.monotonic()
    subprocess.run(cmd, check=True, capture_output=True)
    return time.monotonic() - t0

times = {}

# Issuer: the batch-mint proofs (witness + Groth16 via snarkjs).
for flavor, script in (("b1", "prove_mint_batch.js"),
                       ("a2", "prove_mint_batch_a2.js")):
    args = json.loads((FIX / f"{flavor}.json").read_text())["mint_args"]
    args = [a.replace(f"e2e_{flavor}", f"docflow_{flavor}") for a in args]
    times[f"mint {flavor} (issuer)"] = timed(
        ["node", f"scripts/snark/{script}", *args])

# Depositor: the flavor-agnostic spend proof (against the docflow mint).
payout = json.loads((FIX / "a2.json").read_text())["payout"]
times["spend (depositor, all)"] = timed(
    ["node", "scripts/snark/prove_spend.js",
     "build/snark/mint_batch_a2_n1/fixtures/docflow_a2.json",
     "0", payout, "1", "docflow_a2"])

# Depositor, B1: the repaired membership proof (wasm witness + snarkjs).
b1 = SNARK / "b1_membership"
times["b1 membership (depositor)"] = (
    timed(["npx", "snarkjs", "wtns", "calculate",
          str(b1 / "identity_membership_b1_js/identity_membership_b1.wasm"),
          str(E2E / "b1/b1_membership_input.json"), str(tmp / "b1.wtns")]))
    + timed(["npx", "snarkjs", "groth16", "prove", str(b1 / "b1m_0001.zkey"),
            str(tmp / "b1.wtns"),
            str(tmp / "b1_proof.json"), str(tmp / "b1_public.json")]))

# Depositor, addressed flavors: the folded gate (C++ witness + rapidsnark).
for flavor, nb in (("a1", "deposit_fold_a1"), ("a2", "deposit_fold_a2")):
    gen = SNARK / nb / f"{nb}_cpp" / nb
    zkey = next((SNARK / nb).glob("*_0001.zkey"))
    wit = timed(["bash", "-c",
                f"ulimit -s 65520 && {shlex.quote(str(gen))} "
                f"{shlex.quote(str(E2E / flavor / 'fold_input.json'))} "
                f"{tmp}/{nb}.wtns"])
    prv = timed([str(RAPID), str(zkey), f"{tmp}/{nb}.wtns",
                f"{tmp}/{nb}_proof.json", f"{tmp}/{nb}_public.json"])
    times[f"fold {flavor} (depositor)"] = wit + prv
```

```
for k, v in times.items():
    print(f"{k:30s} {v:6.2f}s")
```

```
mint b1 (issuer)          1.01s
mint a2 (issuer)          0.97s
spend (depositor, all)    0.77s
b1 membership (depositor) 19.73s
fold a1 (depositor)       15.09s
fold a2 (depositor)       17.05s
```

13.3 The bill, per flavor and role

Download = first-run fetch (zkey + witness generator); prove = wall time for every SNARK that party runs for one note, from the measurements above. The artifact-free sigma legs (Schnorr / issuer_reenc / depositor binding / eEnc re-encryption / receipt vd proofs) add milliseconds and nothing to download.

```
BILL = [
    ("issuer    b1/a1 (N=1)", ["mint_batch N=1    (B1/A1 mint)"],
     ["mint b1 (issuer)"]),
    ("issuer    a2    (N=1)", ["mint_batch_a2 N=1  (A2 mint)"],
     ["mint a2 (issuer)"]),
    ("depositor b1",
     ["spend      (all spends)",
      "b1 membership (B1 spend)"],
     ["spend (depositor, all)", "b1 membership (depositor)"]),
    ("depositor a1",
     ["spend      (all spends)",
      "deposit_fold_a1 (A1 spend)"],
     ["spend (depositor, all)", "fold a1 (depositor)"]),
    ("depositor a2",
     ["spend      (all spends)",
      "deposit_fold_a2 (A2 spend)"],
     ["spend (depositor, all)", "fold a2 (depositor)"]),
]
print(f"{party / flavor:24s} {'download+store':>15s} {'prove (per note)':>17s}")
print("-" * 58)
for who, arts, ts in BILL:
    dl = sum(sum(sizes[a]) for a in arts)
    pt = sum(times[t] for t in ts)
    print(f"{who:24s} {dl:>13.1f}MB {pt:>16.2f}s")
```

party / flavor	download+store	prove (per note)
issuer b1/a1 (N=1)	12.7MB	1.01s
issuer a2 (N=1)	14.8MB	0.97s
depositor b1	289.0MB	20.50s
depositor a1	1907.8MB	15.86s
depositor a2	2169.0MB	17.82s

13.4 What this means for the wallet UX

- **Lazy, tiered artifact fetch.** A receive-and-hold wallet needs *zero* SNARK artifacts. A B1-spending wallet needs ~25 MB (`spend` + `identity_membership_b1`). The 1.9 and 2.2 GB fold keys are needed only to *spend an addressed note*, so fetch them in the background on first A1/A2 receipt (the user will plausibly spend later), with hash-pinned, resumable downloads. An issuer wallet sizes its mint key to its batch habit: 10 MB at $N=1$, ~330 MB at $N=32$.
- **One progress bar that matters.** Every proof except the depositor's identity gate is under ~2 s – below UX-noticeable for a payment flow. That gate's ~15-20 s (a fold under rapidsnark,

or B1's membership under `snarkjs`) is the one moment to design for: prove in the background between "note received and verified" and "deposit", or show explicit progress at spend. Proving is local and private – nothing leaves the device – and never blocks the counterparty (the issuer's proof is done before delivery; the depositor's at their leisure).

- **Per-payment chain bandwidth is negligible** next to the artifacts: the depositor replays the commitment tree from `Minted` calldata (32 B per leaf in the pool plus log overhead) and reads one identity-tree path; both are KB-scale at today's pool sizes and cacheable incrementally.
- **Verification is free to distribute**: the Groth16 verifiers are on chain, and the bilateral receipts re-verify with no SNARK artifacts at all – a third party checks a slip with the pure-Python wallet alone.

14 Demurrage-Carrying Transfers

The Note-spend payout in Step 5 went out as a plain `buck.transfer` that BUCK routed through its internal carrying path because the `Notes` pool is bound `isCarrying`. This section unpacks why a single per-account flag – not a separate method, not a marker interface – is sufficient, and what its ledger effect looks like.

14.1 Two Ways to Handle Demurrage at Transfer Time

BUCK demurrage is a balance-time integral: each account stores its `buckSeconds` (the crystallised $\int B dt$) plus the `timestamp` of its last touch, and its accrued fee is `rate \cdot (buckSeconds + balance \cdot elapsed)`. Two ledger responses to that accrued fee are distinguishable at transfer time, selected by the *sender's* `isCarrying` flag:

- *Deducting transfer (sender is non-Carrying, the default)*. On a 1,000-BUCK outgoing, Alice's accrued fee for the pre-transfer balance is settled at Alice – one-shot burned against her balance, and her `buckSeconds` is re-crystallised from the new balance. The 1,000 BUCK then lands at Bob *fresh*: his inflow begins accruing age from now, diluting his effective age. From a system perspective, accrued fees are burned at each transfer and the recipient enters fresh.
- *Carrying transfer (sender is `isCarrying`)*. On the same 1,000-BUCK outgoing, nothing is burned at the sender: the pool sheds a proportional slice of its live buck-seconds (`carried = liveBs \cdot 1000 / poolRaw`) and keeps its per-BUCK age. Bob inherits that slice (`buckSeconds + carried =`), so he absorbs the age of the BUCKs he just received. Nominal balances move by exactly 1,000 on each side; no fee is burned; total system BUCK-age is preserved.

The invariant the carrying variant keeps is: **nominal BUCK out = nominal BUCK in** (no silent burn at the transfer site), with the recipient's forward index shift representing the demurrage they will pay on their next standard outgoing transfer.

14.2 Why the Note Pool Needs It

Three structural features of the pool make the deducting default wrong:

1. **Per-note age opacity.** When Alice mints six notes, the pool records only six opaque Poseidon commitments. The pool cannot deduct "the demurrage *Alice's specific note* owes" at spend time, because the chain does not distinguish one commitment's age from another. Any per-note fee would have to be computed off the pool aggregate, which would force Bob to pay fees that the pool's other depositors owed – and, worse, differentiate one commitment's payout from another, destroying the anonymity set.
2. **Face-to-payout identity.** `face` is the public SNARK output and the value the mint circuit committed to. A deducting pool transfer would deliver `face - fee` to Bob while the proof claimed `face`. This would break the mint-to-spend value-conservation invariant: the SNARK's promise would not match the ERC-20 ledger movement, and the spend circuit would have to become aware of demurrage accounting to compensate.
3. **Pool as bookkeeping buffer.** The pool is not a long-term BUCK holder; every BUCK that enters is earmarked for one future redeemer. Deducting fees at the pool's egress would siphon value into a burn that the pool has no way to replenish, producing a cumulative insolvency as outstanding notes circulate.

The carrying path fixes all three: no per-note age, `face` to payout, no pool insolvency.

14.3 Shipped Surface in `Buck.sol`

There is no separate `transferCarrying` method and no `TransferCarrying` event. The carrying behaviour is selected by a per-account flag, `isCarrying`, set when the account is bound:

```
// IdentityRegistry: governance sets isCarrying at bind time.
function bindContract(address t, ..., bool isPublicIdentity_, bool isCarrying_) external;
mapping(address => bool) public isCarrying;
```

The standard `transfer` path then dispatches on the *sender's* flag:

```
if (identity.isCarrying(from)) _carryingTransfer(from, to, amount);
else _nonCarryingTransfer(from, to, amount);
emit Transfer(from, to, amount); // one standard event either way
```

The preconditions are exactly those of any transfer (sender and recipient both `isVerified`, the bilateral receipt-fragment / `_identityHash` identity check), because it *is* an ordinary transfer. `_carryingTransfer` sheds a proportional slice of the sender's live buck-seconds onto the recipient rather than burning a fee at the sender:

```
liveBs    = from.buckSeconds + from.raw * elapsed;
carried   = liveBs * amount / from.raw;
from.buckSeconds    = liveBs - carried; // sender keeps its per-BUCK age
recipient.buckSeconds += carried;      // recipient inherits the age
```

Carrying accounts hold no NFT-backed credit and cannot go negative (the path asserts `amount < raw`). What makes an account pool-friendly is the *flag*, set once at bind time; there is no per-call method to invoke and no balance precondition beyond that non-negative assertion.

14.4 Who Else Is Carrying

The Note pool is the canonical `isCarrying` account, but the flag is not pool-private: governance binds AMM pools, the Jubilee, and the Notes pool `isCarrying`, and could bind any transitory-holder contract the same way – escrow accounts whose beneficiaries are the true owners, payroll contracts pulling from an employer, cross-chain bridge egresses. The semantic invariant is "the account is a transitory holder of BUCK earmarked for its recipients," and in that case carrying the age forward matches the economic reality.

For ordinary peer-to-peer accounts the deducting default remains correct: the sender is paying from their own pocket and is the natural place to burn their own accrued fee, so EOAs are bound non-Carrying.

14.5 Open Question: Identity Receipts on Carry

Because a carrying transfer *is* an ordinary `transfer`, it uses the same identity predicate – both parties `isVerified`, with the receipt-fragment fallback to `_identityHash` on the public side when a party has `isPublicIdentity = true`. The Notes pool side relies on its bound Public Identity to satisfy the bilateral check. Whether a carrying transfer between two Encrypted-Identity counterparties should require its own per-call Chaum-Pedersen receipt is an open question; for the Notes pool it is moot (the pool is bound under a Public Identity), but if transitory-holder carry is ever opened to private-to-private pairs an approve-style receipt may be wanted for auditability. See the **Open Work Items** in alberta-buck-ethereum.org.

15 Putting the Guarantees Side-by-Side

The mint walkthrough established five guarantees; the spend walkthrough adds four more that become concrete only once the redemption path exists.

Stage	Guarantee	Mechanism
Mint	Atomic conservation	Single EVM call: proof verify, <code>transferFrom</code> , append-cms all-or-nothing
Mint	Issuer-bound, recipient-blinded	<code>msg.sender</code> is public; <code>cm_i</code> is Poseidon-hidden over (<code>flavor</code> , <code>v</code> , <code>rho</code> , <code>idHash</code> , <code>pred</code>)
Mint	Per-note disclosure (recipient only)	Off-chain opening contains exactly one note's witness fields
Mint	Offline receipt reconstruction	<code>T_mint</code> calldata + opening + on-chain Identity lookup is sufficient and self-checkable
Mint	Adversarial-prover soundness	Groth16 knowledge soundness on fixed on-chain verifier; ERC-20 balance/allowance integrity
Spend	Redemption unlinkable to specific cm	<code>cm</code> is a <i>private</i> witness under public <code>noteRoot</code> ; only nullifier + face + recipient revealed
Spend	Double-spend prevented	Deterministic nullifier + one-bit <code>nullifiers</code> mapping; rejected on reuse
Spend	Recipient-bound payout	<code>recipient</code> is a public SNARK input; proof rejects if re-targeted in-flight
Spend	Value conservation across mint<->spend	Range-bounded mint $v[i] + \text{spend face} == v$; cumulative payouts $\leq \text{totalFace}$ per batch

All nine rows are enforced by code in the tree. On the mint side the Merkle insertion lives *inside* the SNARK (`mint_batch.circom`, per-N Groth16 verifiers, per-N adapter dispatch); the chain writes the SNARK-attested `newRoot` into the 30-slot root-history ring `Notes.sol` maintains for the spend side. On the spend side: the flavor-agnostic `spend.circom` note proof, `SpendGroth16Verifier`, `SpendVerifierAdapter`, the on-chain Merkle accumulator, and BUCK's `isCarrying` carrying-transfer path. The IdentityRegistry provides the issuer-attestation and recipient-verification legs. Addressed spends verify one folded Groth16 proof. B1 verifies `verifyDepositorBinding` and `identity_membership_b1`. Theorems 7-12 of the formal proofs are the argument for these gates. The wallet cost of the fold is in *The Prover Bill of Materials* below: fifteen to eighteen seconds and a 1.9 to 2.2 GB proving key per addressed note, measured on this machine.

Why the spend carries this apparatus at all – anonymity is the cost. A direct transfer reaches the same mutual-decryptability guarantee with no SNARK, because the bilateral `approve`

names both parties at once. The Notes detour pays the mint SNARK, the recipient-key blinding, and the folded gate purely to buy mint $\leftrightarrow$ spend unlinkability: the issuer's registered anchor lives at mint, the recipient's at spend, and they are never co-present, so the apparatus is what bridges them. Public-issuer notes (A1/B1) keep the cheap, fully-sound path; A2 is the premium that hides the issuer too. See *alberta-buck-notes* ("Why A2 Costs More: Anonymity Is the Cost") and the privacy audit below ("The EOA Transfer Mirror").

16 Deployment Status and Open Questions

The redemption walk above is the identity-aware spend. Addressed notes use the folded gate when `depositFoldVerifier` is set. B1 uses `verifyDepositorBinding` and `identity_membership_b1.circom` when that adapter is the membership slot. What is actually in the tree:

1. **On-chain identity root.** Governance appoints a root authority, which enrolls each registry's subtree at an aggregator slot and appoints the aggregator; the aggregator alone posts roots, into a ring of 256 with their times, and each consumer declares the oldest it accepts (Notes: seven days). A receiving leaf is admitted by its registry: `register` and `bindContract` refuse a caller-supplied leaf, and must, since a call that accepted any leaf would accept one pairing any Identity with any key. The organisation admits the leaf to its own subtree, the aggregator composes, and a membership path is 32 levels. The canonical off-chain trees are `alberta_buck.registry.tree` and `merkle_service`.
2. **B1 membership.** `identity_membership_b1.circom` proves $T = bH_{\text{PEDERSEN}}$, the salted leaf, and the path. `IdentityMembershipB1VerifierAdapter` is the verifier, and it is the only membership adapter in the tree.
3. **Addressed spends.** `spendCoupledA1` `spendCoupledA2` verify one proof. `DepositFoldVerifierAdapter` derives the public inputs and reads the depositor's registered account. The slot cannot be cleared and an addressed spend without it reverts, so there is no configuration in which a weaker gate answers for one. `spendCoupledB1` composes its sigma with the membership verifier, passing P_{dep} into it – sound there because B1's two facts rest on one secret.
4. **The note tie lives in the fold.** `spend.circom` exposes no `idHash`. `deposit_fold_a1` and `deposit_fold_a2` recompute the nullifier from the `idHash` they open, so the ciphertext they check is this note's.
5. **The receipt names by evidence, not by derivation.** The canonical AB-RCPT verifier no longer derives the payee's identity scalar and decrypts the addressed ciphertexts with it: that procedure worked for anyone who had ever seen a receipt. The recipient proves by verifiable decryption under pk_{recv} ; the issuer, which cannot open its own ciphertexts, discloses the mint randomness instead; and the mailbox leaf ties that key to the named Identity with a Poseidon and a path. An issuer-side receipt without it prints **UNVERIFIED RECIPIENT**. An A2 receipt also checks $C_i - T + \gamma H_{\text{PEDERSEN}}$ against the Identity it names.
6. **The payload's scalars are wrapped to the mailbox key.** A delivery must carry r' (the fold needs the SCALAR $r'k$, which the point kR does not give) and, for A2, the issuer's naming salt and T 's blind γ . In the clear each undoes something it travels beside – r' turns *eIss* into plaintext, the salt turns the issuer's naming leaf into a scannable one, γ turns T into $r'pk_{\text{recv}}$ and so the issuer's Identity. All travel masked by `keccak(WRAP_DOMAIN||field||S)`,

with $S = r' \cdot pk_{\text{recv}} = k \cdot R$: Diffie-Hellman, so only the addressee unwraps, and the minter needs no round trip to wrap. One mask per field, because one pad over two scalars leaks their difference.

The end-to-end test spends all three flavours through those gates, with dev-entropy setups. A production ceremony has not been run.

Additional items tracked across the series:

- **Mint-side deferred-approve handshake (shipped).** The public-issuer Schnorr binding (`verifyIssuerSchnorr`) and the private-issuer re-encryption binding (`verifyIssuerReenc`) are deployed and tested. Every mint carries a verified issuer Identity binding.
- **Variable-N batch size (resolved via pinned set).** The wallet rounds any requested split up to the smallest pinned $N \geq N_{\text{requested}}$ from the deployed set (recommended $\{16, 128, 1024\}$), padding with $v = 0$ dummy openings. Each pinned N has its own verifier contract; `MintVerifierAdapter` dispatches on `cms.length`.
- **Prover contention (resolved operationally, not protocolly).** Batch minting makes each mint a rollup-style transaction with a stale-state guard on `oldRoot` and `nextLeafIndex`. Losers in a race re-prove; no BUCK is lost. Initial deployment uses wallet-side optimistic retry; a sequencer/aggregator is deferred until real-world contention warrants it.

17 Privacy Audit: Data Revealed Per Step

The whole construction threads one needle: publish enough that each party can *name* the other under compulsion, while publishing so little that no *third party* can name either, or link payments in bulk. This section consolidates the per-step disclosure analysis for the shipped Identity-M surface.

17.1 Requirements and adversaries

Three properties, against three adversaries:

- **(Bulk-tracing resistance.)** *Mallory*, a passive observer with the full chain history and unbounded compute but no party's secrets, cannot reconstruct who paid whom, how much, or link a mint to its spend(s).
- **(Third-party non-disclosure.)** Nothing on chain reveals either party's KYC-bound Identity point M to a non-participant – not even to the *counterparty's* counterparty, and not in aggregate across many payments.
- **(Mutual decryptability.)** Each of the two participants, from data they hold plus public chain state, can produce a sound receipt naming the other's registered M – and a *colluding* pair cannot arrange a payment for which no such receipt exists (the non-deniable-receipt invariant).

The first two are *confidentiality* against outsiders; the third is *accountability* between insiders. They pull in opposite directions, so every published datum below is justified against all three.

17.2 The on-chain footprint of each step

Step	Published on chain
register / bindContract	addr, pk, E_addr (ElGamal ct of M under own pk), isPublicIdentity, identityRoot
mint (any flavor)	Minted(issuer, totalFace, startIndex, count, newRoot); cms[] in calldata (opaque Poseidon)
mint public (A1/B1)	+ IssuerBound(issuer, newRoot); one Schnorr over keccak(cms)
mint private (A2)	+ IssuerReencBound(issuer, newRoot, count); per-leaf $eIss$, T and the blinded binding ($Q, U, A_1..A_5$)
A1/A2 spend	nullifier, face, recipient, the identity root, $eEnc$, one fold proof
B1 spend	SpentCoupledB1(nf, face, recipient, issuer, eDepForIss); the identity root, P_{dep} , the depositor-binding sigma

Two facts do the heavy lifting throughout. First, cms[] are Poseidon commitments: a leaf reveals *nothing* about ($flavor, v, \rho, idHash, predicate$) without ρ . Second, the spend nullifier $nf = \text{Poseidon}_3(\rho, idHash, 4242)$ is a PRF output over secret material, so it is unlinkable to any cm without the opening (*Proofs*, Theorem 10).

17.3 Why bulk-tracing and third-party privacy hold

Mint is opaque; spend is unlinkable to mint. At mint Mallory sees opaque cms[] and a totalFace; she cannot read any v_i , recipient, or flavor (Poseidon hiding). At spend she sees a fresh nf , a face, and a recipient; nf is a PRF over $(\rho, idHash)$, so inverting it to a specific cm is pre-image resistance. No field of a spend points back to a mint: the spend's $eEnc$ is a *fresh re-encryption* of the leaf ciphertext (uniformly re-randomised; the fold deliberately proves re-encryption *equivalence*, never equality with the public mint-time $eIss$, and T never appears at spend). The only residual linkage surfaces are the two that *must* be public for the ERC-20 to pay out – face and recipient – mitigated by fixed denominations and relayed payouts, orthogonal to this document.

The stronger adversary: someone who already holds an Identity. The arguments above are stated against Mallory, who holds no secrets. The interesting adversary holds several: an issuer knows every Identity scalar it certified, and every past counterparty knows one. Their scan is "is any note in existence addressed to anyone on my list?", and it is defeated by *where the addressing key comes from* rather than by any of the arguments above. Because a note is keyed to pk_{recv} and never to the Identity point, a candidate Identity tests nothing: the residual question is a decisional Diffie-Hellman decision under a key the holder has disclosed to nobody. *The Receiving Key* above shows the scan running, and finding nothing.

Identities never appear on chain – the load-bearing check. A registered E_{addr} is an ElGamal ciphertext of M ; reading it yields nothing without sk . The danger is in the *bindings*, which manipulate M and the recipient point algebraically. Concretely, the A2 mint binding must publish enough to prove " $eIss$ re-encrypts the issuer's registered M_{iss} under the recipient's point" without revealing either. This is exactly where the first draft failed – the canonical near-miss, kept here because it is the clearest way to see the invariant:

Near-miss (rejected draft). The binding published $T = r' * pk_{rec}$.

But the leaf is $C_i = M_{iss} + r' * pk_{rec}$, so ANY observer computes

$$M_{iss} = C_i - T.$$

At mint msg.sender IS the issuer, so this publicly bound the issuer's address to its Identity -- collapsing A2 to A1 and breaking BOTH third-party non-disclosure and bulk unlinkability (every A2 mint by the same issuer now carries the same recoverable M_{iss}).

The shipped binding blinds it: it publishes $T = r'pk_{rec} + \gamma H_{\text{PEDERSEN}}$ for a fresh γ . Now $C_i - T = M_{iss} - \gamma H_{\text{PEDERSEN}}$, a uniformly random offset from M_{iss} – so T reveals nothing to

anyone without γ , which only the recipient receives. The recipient’s key is likewise hidden (inside the perfectly-hiding Q and the uniform U, T), and soundness is undisturbed because γ cancels in the verification relations. The blind is on the hash-to-curve generator rather than G because the fold opens T again: a blind whose logarithm is known would absorb any difference of Identities. An A2 mint’s on-chain footprint is, to Mallory, a vector of uniformly-distributed group elements.

The spend-side proofs reveal the spender’s action, not any Identity. The folds are zero-knowledge; their public inputs are the nullifier (PRF), $eEnc$ (uniform), the identity root, and the depositor’s own registered key and credential. B1’s depositor-binding sigma is HVZK and its only Identity-derived public value is $P_{dep} = M_{dep} + bH$, perfectly hiding for uniform b . `msg.sender` is visible – exactly as the sender of any ERC-20 transfer is – but no *Identity* is.

17.4 Why mutual decryptability holds – the final data set

For each flavor, the data each party ends up holding, and the predicate it satisfies (the receipt verifiers in `alberta_buck.wallet` are the public predicate `RcptVerify`; see the executable sections above):

Flavor	Recipient names issuer via	Issuer names depositor via
B1	in-clear M_{iss} + batch Schnorr (<code>verifyIssuerSchnorr</code>)	SpentCoupledB1 : decrypt e
A1	in-clear M_{iss} + batch Schnorr	chose M_{rec} at mint; the m
A2	decrypt $eIss$ under k ; binding, key tie and membership prove it is the registered minter (A2 in Code)	chose M_{rec} at mint; the m

The A2 row is the subtle one. The recipient’s receipt composes four facts: the mint anti-framing binding (the ciphertext carries the issuer’s *registered* record), their own verifiable decryption (it opens under pk_{recv} to the named point), the key tie $C_i - T + \gamma H_{\text{PEDERSEN}} = M_I$ (the binding’s Identity is the one decrypted), and the membership of that point in rt_{id} . No issuer cooperation is needed – the receipt is *unilateral* – and no new proof is needed at receipt time.

17.5 The colluding pair, and where each escape is closed

Completeness (an honest recipient *can* produce a receipt) is met off chain. The third clause – no deniability even when payer and payee *collude* – needs the binding to be *unavoidable*. Three escapes existed on the road here, each now closed at a specific gate:

1. *A binding-less leaf.* Closed at **mint**: `Notes.mint` requires a verified issuer binding (Schnorr or re-encryption) for every leaf; the SNARK’s per-leaf `issuerMode` signal makes a private-issuer bearer leaf unprovable (no B2 codepoint).
2. *A nameable note deposited by the wrong Identity.* Closed at **spend** by the fold: the depositing account’s Identity and the key that reads the note must be paired in one certified leaf (*Proofs*, Theorem 8).
3. *An un-nameable note made spendable by ciphertext substitution.* Closed at **spend** by the note tie inside the fold: the verified $eEnc$ must be keyed to the identity material committed behind *this spend’s nullifier* (A2: re-encrypt the committed $eIss$; A1: share the $eNote$ ’s key, the public **face** pinning it) (*Proofs*, Theorem 8) – demonstrated executably in the A2 and A1 in Code counter-examples above.
4. *A misnamed note: a registered puppet in place of the minter.* Closed at **spend** by the key tie: the note commits the binding’s T and the A2 fold opens it against the recipient’s own key (*Proofs*, Theorem 12) – the split mint above is refused by the delivery, the fold and the receipt.

18 Security Properties and the Tests That Witness Them

The Notes security and identity model (all three flavours with Merkle membership) is shipped and exercised end-to-end – the on-chain lifecycle is measured by `test/NotesE2E.t.sol` (see *alberta-buck-notes* "On-Chain Gas"). This section records the shipped surface, the remaining integration, and the document series.

Component	Source
Notes contract: roots, nullifiers, mint + spendCoupled{A1,A2,B1}	<code>src/Notes.sol</code>
Carrying-transfer path in BUCK	<code>src/Buck.sol</code> (<code>_carryingTransfer</code> , <code>buckSeconds</code> apportionment)
Mint Groth16 circuits + per-N verifiers (PUBLIC + PRIVATE modes)	<code>circuits/mint_batch.circom</code> , <code>circuits/mint_batch_a2.circom</code> , <code>src/circuits/deposit_fold_a1.circom</code> , <code>deposit_fold_a2.circom</code> , <code>Deposit</code>
A1/A2 folded deposit gate	<code>IdentityRegistry.verifyDepositorBinding</code> , <code>alberta_buck.wallet.b</code>
B1 depositor binding (EIP-196 Okamoto sigma, A2 dual)	<code>IdentityRegistry.verifyIssuerReenc</code> , <code>test/IssuerReenc.t.sol</code>
A2 issuer re-encryption binding (anti-framing)	<code>IdentityRegistry.verifyIssuerSchnorr</code> , <code>test/IssuerSchnorr.t.sol</code>
Public-issuer Schnorr binding (A1/B1 mint)	<code>circuits/identity_membership_b1.circom</code> , <code>IdentityMembershipB1Ve</code>
B1 membership circuit (salted leaf, H_{PEDERSEN})	<code>alberta_buck/registry/</code> , <code>test/test_registry.py</code> (49)
Identity registry prototype (certificates, trees, aggregator)	<code>scripts/gen_unilateral_a2_vectors.py</code> , <code>scripts/gen_b1_binding_v</code>
Python Solidity cross-artifact vectors	<code>alberta_buck/sim/identity.py</code>
Registry-backed sim identity issuance	<code>src/PoseidonT3Bytecode.sol</code>
Poseidon-T3 precompile (in-EVM bytecode)	<code>src/MintVerifierAdapter.sol</code> , <code>src/MintVerifierA2Adapter.sol</code> , <code>src</code>
Verifier adapters (proof plumbing, per-N mint dispatch)	<code>Notes.spendCoupled{A1,A2,B1}</code> , <code>alberta_buck.wallet.unilateral_a</code>
Identity-M-bound spend entrypoints (A1/A2/B1) + e2e tests	<code>scripts/snark/prove_*</code> , <code>scripts/snark/setup.sh</code>
Prover scripts (witness gen + pta ceremony)	

Every property claimed in this document is pinned by an executable witness – the Python suites exercise the wallet reference, the Forge suites the on-chain verifiers, and the cross-artifact vectors (`alberta_buck.wallet` → `vm.readFile`) keep the two byte-for-byte aligned.

Suite	Count	What it demonstrates
<code>alberta_buck/test/test_unilateral_a2.py</code>	17	A2 mint binding; delivery decrypt; the fold from any account; unilateral receipt names
<code>alberta_buck/test/test_unilateral_a1.py</code>	12	A1 mint; <code>idHash</code> binds the public issuer + Schnorr; recipient-side receipt names both; t
<code>alberta_buck/test/test_b1_binding.py</code>	11	B1 depositor binding; issuer-unilateral receipt names the depositor; collusion-substituti
<code>alberta_buck/test/test_e2e_payload.py</code>	7	the delivered payload SUFFICES: the shipped randomness opens the shipped ciphertex
<code>test/B1Binding.t.sol</code>	6	on-chain <code>verifyDepositorBinding</code> parity + soundness
<code>test/NotesCoupledB1.t.sol</code>	8	end-to-end <code>spendCoupledB1</code> : the depositor-binding + bound membership path
<code>test/NotesA2Tie.t.sol</code>	4	the A2 mint commits T : the binding and the mint proof must agree on it
<code>test/IdentityAccumulator.t.sol</code>	10	the root authority, subtree enrollment, the root ring and consumer ages, public paths t
<code>test/NotesIdentityMembership.t.sol</code>	6	the gate slots: the fold verifier cannot be cleared, and an addressed spend without it re
<code>test/NotesE2E.t.sol</code>	12	the full lifecycle per flavour with EVERY verifier real (batch mint, spend, and the fold

Build/test entry points: `make nix-test` (Forge), `make nix-test-python` (wallet reference), `make nix-snark-e2e-fixtures` (the all-real-verifier lifecycle fixtures).

19 Cross-References

- alberta-buck-notes.org – architectural overview of the Notes primitive (flavors, Merkle tree, demurrage interaction). The other living half of the Notes documentation pair; this document is the worked, executable half.
- alberta-buck-proofs.org – formal correctness arguments for these gates (Part IV, Theorems 7-12).
- alberta-buck-verifier-implementation.org – SNARK toolchain engineering: the witness-generation root cause, rapidsnark integration, trusted setups.
- alberta-buck-ethereum.org – Solidity surface, gas costs, phase-by-phase implementation status, Open Work Items.
- alberta-buck-identity.org – IdentityRegistry, PS credentials, and the bilateral ElGamal handshake that anchors `msg.sender` to a verifiable identity.
- The unified identity-M / "one gadget" / Registry-Identity accumulator design (commit-before-use) is now in alberta-buck-notes.org ("Mutual Decryptability", "one gadget") and here in "The Identity-M Spend Path" + accumulator sections. (Superseded the original memo.)
- Non-deniable-receipt invariant: see alberta-buck-notes.org "The Non-Deniable-Receipt Invariant"; per-step privacy audit and "EOA Transfer Mirror" are above in this document; theorems in alberta-buck-proofs.org. The mint-side issuer bindings are wallet sigmas (`verifyIssuerSchnorr`, `verifyIssuerReenc`); the note $\leftrightarrow$ `eEnc` tie is a relation inside the folded gate (see "A2 Identity-Targeted Spend").
- `circuits/mint_batch.circom` – the batch-mint circuit (one compile per pinned N).
- `circuits/spend.circom` – the flavor-agnostic spend (note-proof) circuit.
- `circuits/deposit_fold_a1.circom`, `circuits/deposit_fold_a2.circom` – the addressed gate: one witness, four relations for A1 and six for A2, including the note $\leftrightarrow$ `eEnc` tie and A2's key tie.
- `circuits/identity_membership_b1.circom` – B1's membership, with its blind proven rather than witnessed on a generator of unknown logarithm.
- `src/Notes.sol`, `src/MintGroth16Verifier*.sol`, `src/MintVerifierAdapter.sol`, `src/SpendGroth16VerifierAdapter.sol`, `src/DepositFoldA1Verifier.sol`, `src/DepositFoldA2Verifier.sol`, `src/DepositFoldVerifierAdapter.sol`, `src/IdentityMembershipB1Verifier.sol`, `src/IdentityMembers...` – the on-chain code paths walked above.
- `scripts/snark/prove_mint_batch.js`, `scripts/snark/prove_spend.js` – the reference provers used in tests; `scripts/snark/setup_deposit_fold.sh`, `scripts/snark/setup_b1_membership.sh` – the gate circuits + trusted-setup pipelines.
- `test/MintVerifier.t.sol`, `test/SpendVerifier.t.sol`, `test/NotesE2E.t.sol` – the Foundry tests that pin the soundness stories to executable assertions.